$$\frac{d}{d\,intercept}$$ Sum of squared residuals =

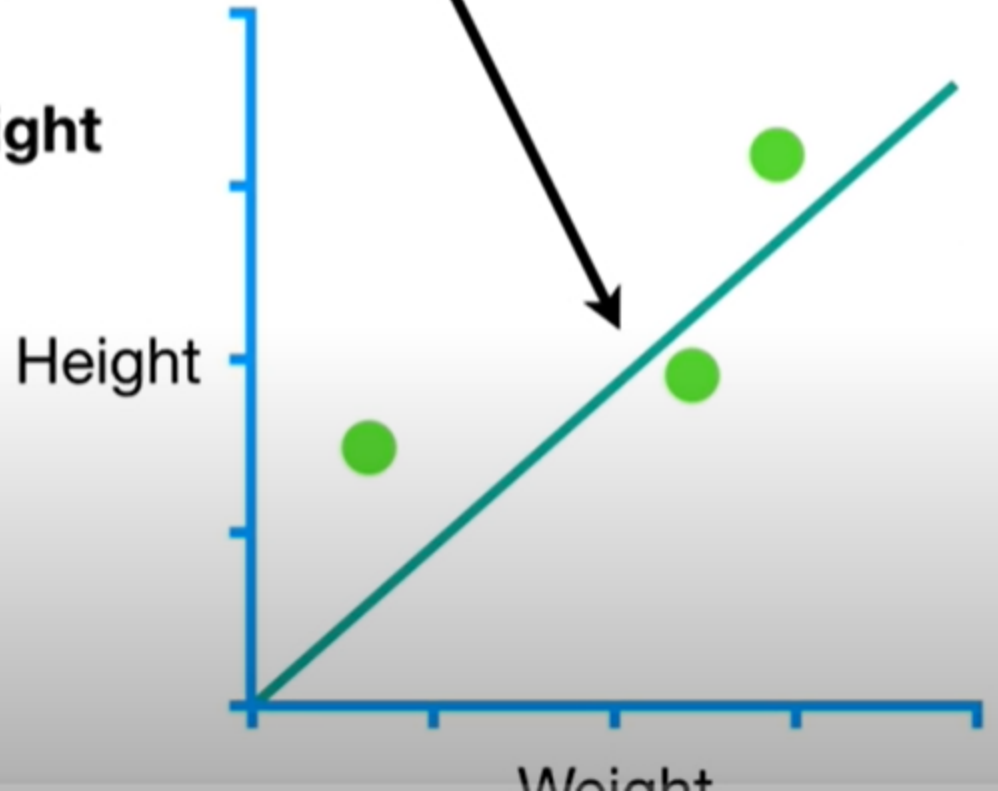
$$-2(\mathbf{1.4} - (\text{intercept} + \text{slope} \times \mathbf{0.5}))$$

$$+ -2(\mathbf{1.9} - (\text{intercept} + \text{slope} \times \mathbf{2.3}))$$

$$+ -2(\mathbf{3.2} - (\text{intercept} + \text{slope} \times \mathbf{2.9}))$$

In this super simple example, we were just fitting a line with two parameters, the **intercept** and the **slope**…

Predicted Height = intercept + slope × **Weight**

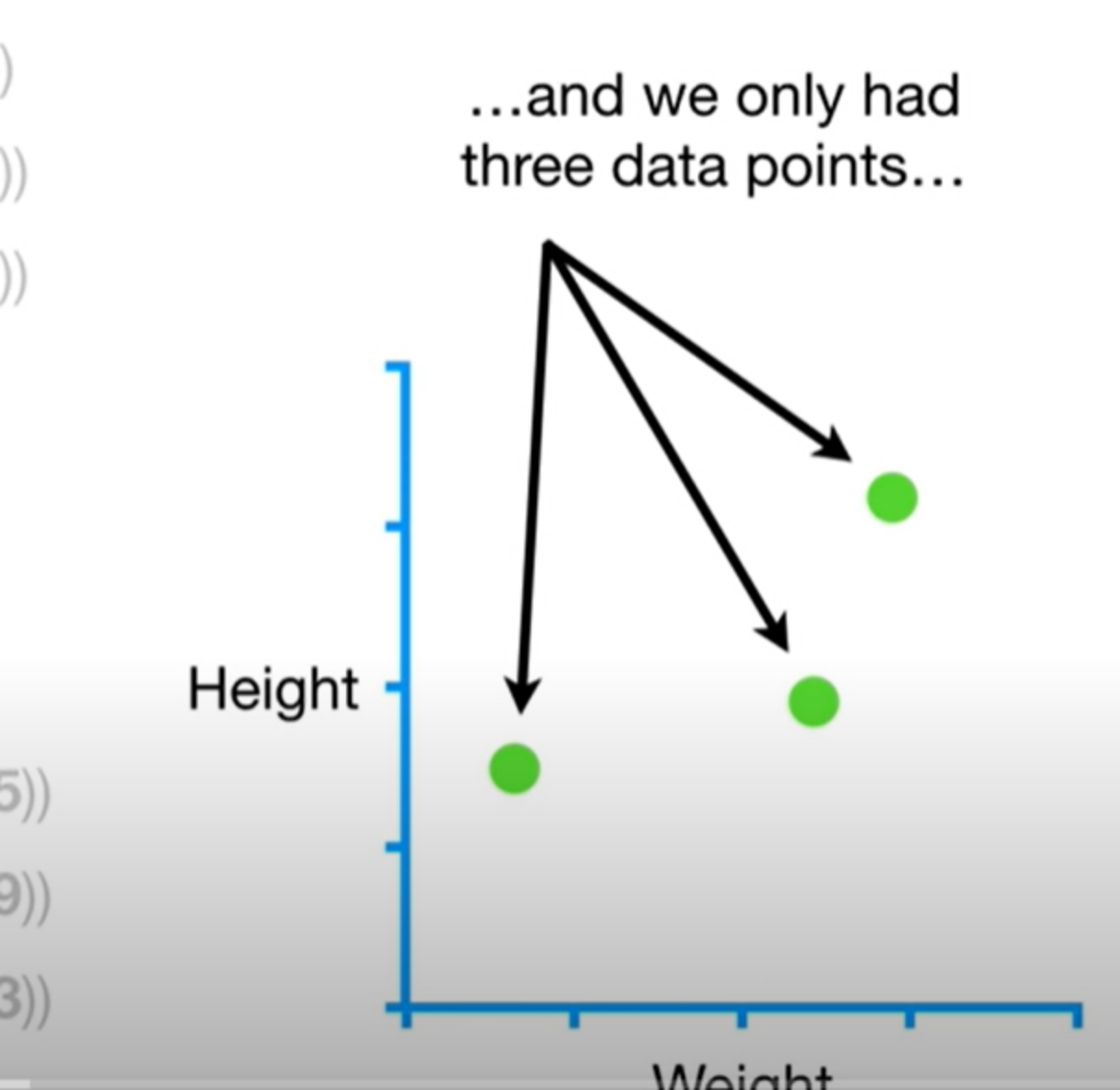


$$\frac{d}{d\,slope}$$ Sum of squared residuals =

$$-2 \times \mathbf{0.5}(\mathbf{1.4} - (\text{intercept} + \text{slope} \times \mathbf{0.5}))$$

$$+ -2 \times \mathbf{2.9}(\mathbf{3.2} - (\text{intercept} + \text{slope} \times \mathbf{2.9}))$$

$$+ -2 \times \mathbf{2.3}(\mathbf{1.9} - (\text{intercept} + \text{slope} \times \mathbf{2.3}))$$


…and we only had
three data points…
Height
Weight

$$\frac{d}{d\ intercept}$$ Sum of squared residuals =

$$-2(\mathbf{1.4} - (\text{intercept} + \text{slope} \times \mathbf{0.5})$$

$$+ -2(\mathbf{1.9} - (\text{intercept} + \text{slope} \times \mathbf{2.3}))$$

$$+ -2(\mathbf{3.2} - (\text{intercept} + \text{slope} \times \mathbf{2.9}))$$

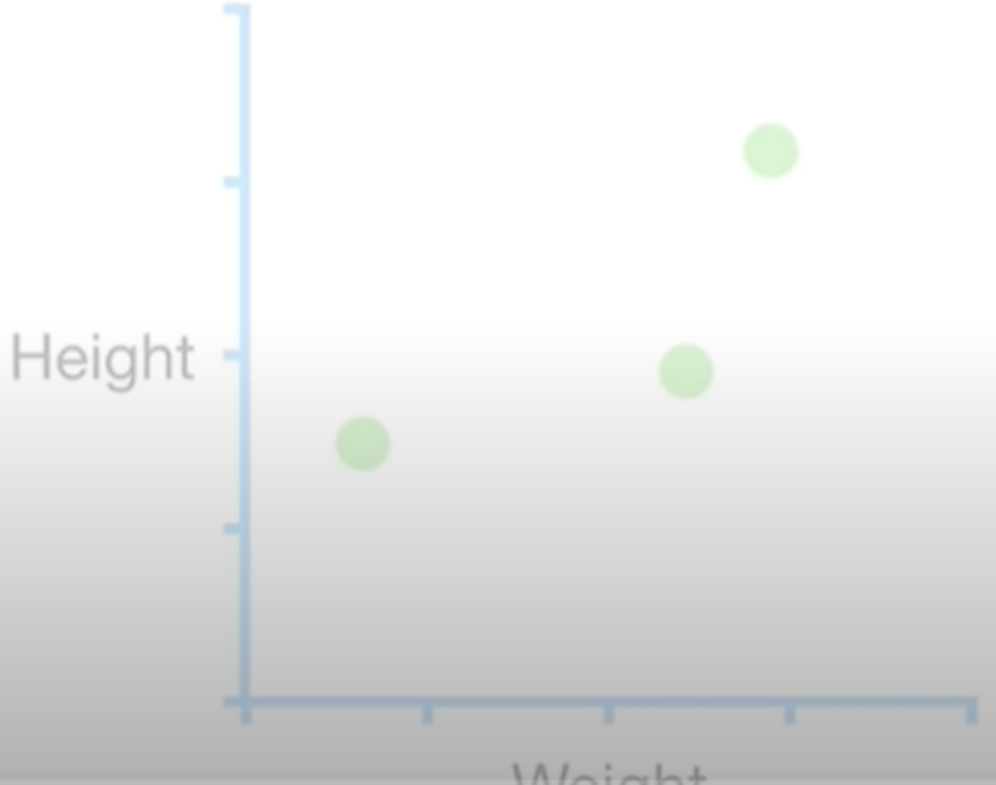
…so we only had three terms to compute each step for the **intercept**…

$$\frac{d}{d\ slope}$$ Sum of squared residuals =

$$-2 \times \mathbf{0.5}(\mathbf{1.4} - (\text{intercept} + \text{slope} \times \mathbf{0.5}))$$

$$+ -2 \times \mathbf{2.9}(\mathbf{3.2} - (\text{intercept} + \text{slope} \times \mathbf{2.9}))$$

$$+ -2 \times \mathbf{2.3}(\mathbf{1.9} - (\text{intercept} + \text{slope} \times \mathbf{2.3}))$$



$$\frac{d}{d\ intercept}$$ Sum of squared residuals =

-2(**1.4** - (intercept + slope × **0.5**)

+ -2(**1.9** - (intercept + slope × **2.3**))

+ -2(**3.2** - (intercept + slope × **2.9**))

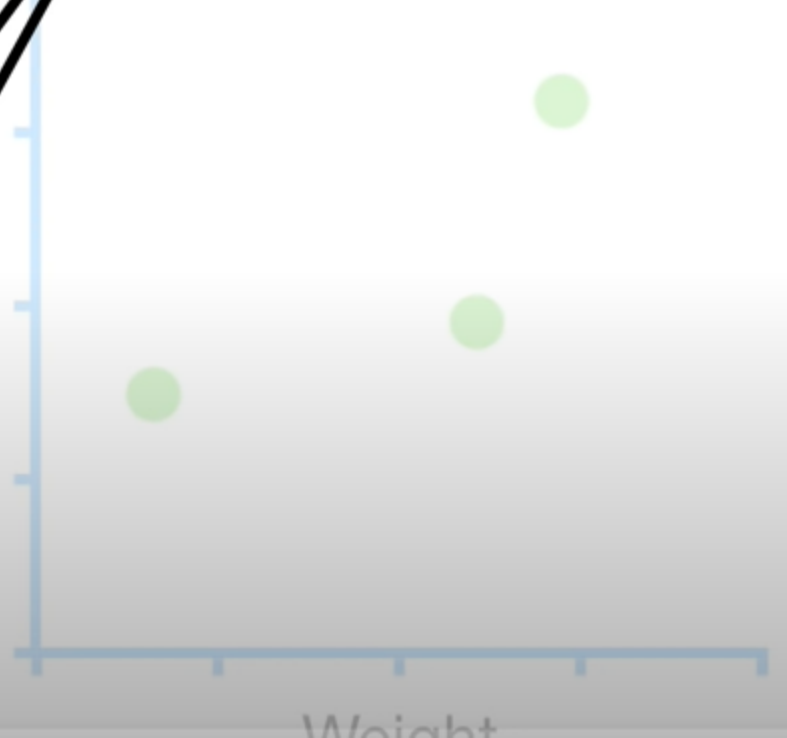
…and we only had three terms to compute each step for the **slope**…

$$\frac{d}{d\ slope}$$ Sum of squared residuals =

-2 × **0.5**(**1.4** - (intercept + slope × **0.5**))

+ -2 × **2.9**(**3.2** - (intercept + slope × **2.9**))

+ -2 × **2.3**(**1.9** - (intercept + slope × **2.3**))



…so each step didn’t require that much math.

But what if we had a more complicated model, like a **Logistic Regression** that used **23,000** genes to predict if someone will have a disease?

$\frac{d}{d\ gene1}$ Loss Function()

$\frac{d}{d\ gene2}$ Loss Function()

$\frac{d}{d\ gene3}$ Loss Function()

$\frac{d}{d\ gene4}$ Loss Function()

$\frac{d}{d\ gene5}$ Loss Function()

$\frac{d}{d\ gene6}$ Loss Function()

$\frac{d}{d\ gene7}$ Loss Function()

Then we would have
23,000 derivatives to plug
the data into.

$$\frac{d}{d\,gene1}\,\text{Loss Function()}$$

$$\frac{d}{d\,gene2}\,\text{Loss Function()}$$

$$\frac{d}{d\,gene3}\,\text{Loss Function()}$$

$$\frac{d}{d\,gene4}\,\text{Loss Function()}$$

$$\frac{d}{d\,gene5}\,\text{Loss Function()}$$

$$\frac{d}{d\,gene6}\,\text{Loss Function()}$$

$$\frac{d}{d\,gene7}\,\text{Loss Function()}$$

And what if we had data
from **1,000,000** samples?

$$\frac{d}{d\ gene1}$$ Loss Function()

$$\frac{d}{d\ gene2}$$ Loss Function()

$$\frac{d}{d\ gene3}$$ Loss Function()

$$\frac{d}{d\ gene4}$$ Loss Function()

$$\frac{d}{d\ gene5}$$ Loss Function()

$$\frac{d}{d\ gene6}$$ Loss Function()

$$\frac{d}{d\ gene7}$$ Loss Function()

Then we would have to calculate **1,000,000** terms for each of the **23,000** derivatives.

$\frac{d}{d\,gene1}$ Loss Function()

$\frac{d}{d\,gene2}$ Loss Function()

$\frac{d}{d\,gene3}$ Loss Function()

$\frac{d}{d\,gene4}$ Loss Function()

$\frac{d}{d\,gene5}$ Loss Function()

$\frac{d}{d\,gene6}$ Loss Function()

$\frac{d}{d\,gene7}$ Loss Function()

Then we would have to calculate **1,000,000** terms for each of the **23,000** derivatives.

In other words, we'd have to calculate **23,000,000,000** terms for each step.

And since it is common to take at least **1,000** steps, we would calculate at least **2,300,000,000,000** terms.

$$\frac{d}{d\,gene1}\,\text{Loss Function()}$$

$$\frac{d}{d\,gene2}\,\text{Loss Function()}$$

$$\frac{d}{d\,gene3}\,\text{Loss Function()}$$

$$\frac{d}{d\,gene4}\,\text{Loss Function()}$$

$$\frac{d}{d\,gene5}\,\text{Loss Function()}$$

$$\frac{d}{d\,gene6}\,\text{Loss Function()}$$

So, for **BIG DATA**, **Gradient Descent** is **slow**.

$$\frac{d}{d\ gene1}\ \text{Loss Function()}$$

$$\frac{d}{d\ gene2}\ \text{Loss Function()}$$

$$\frac{d}{d\ gene3}\ \text{Loss Function()}$$

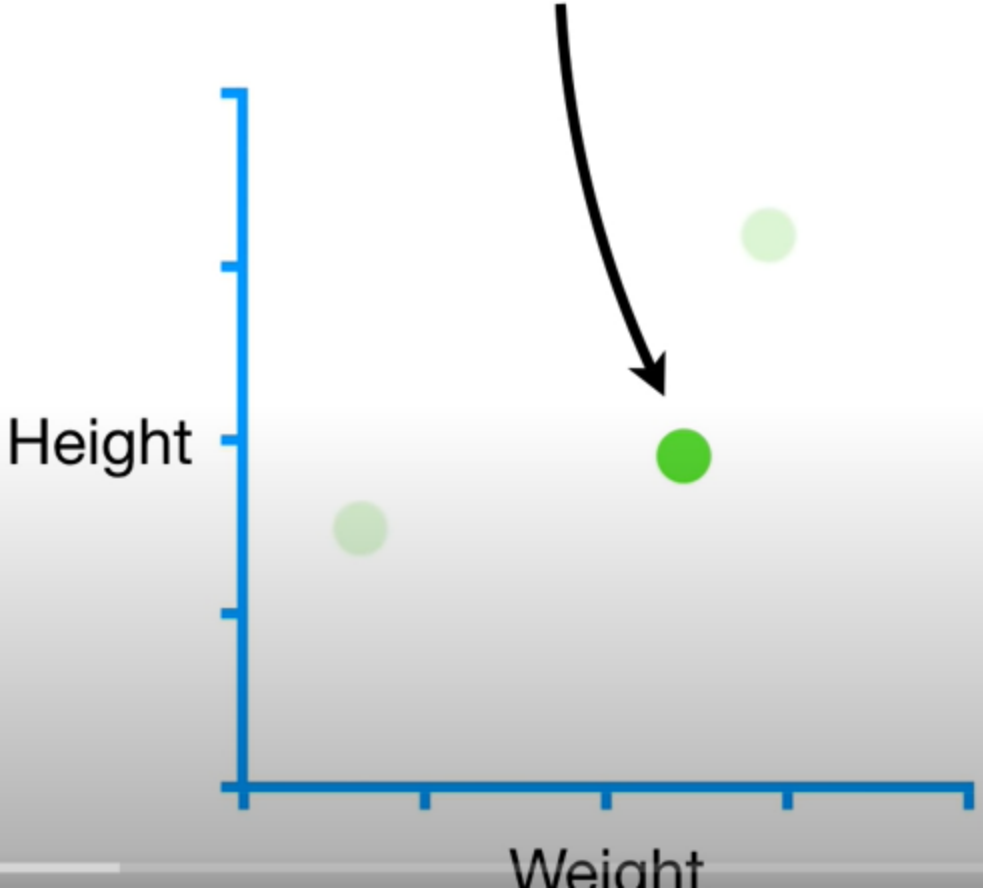
$$\frac{d}{d\ gene4}\ \text{Loss Function()}$$

$$\frac{d}{d\ gene5}\ \text{Loss Function()}$$

$$\frac{d}{d\ gene6}\ \text{Loss Function()}$$

This is where **Stochastic Gradient Descent** comes in handy.

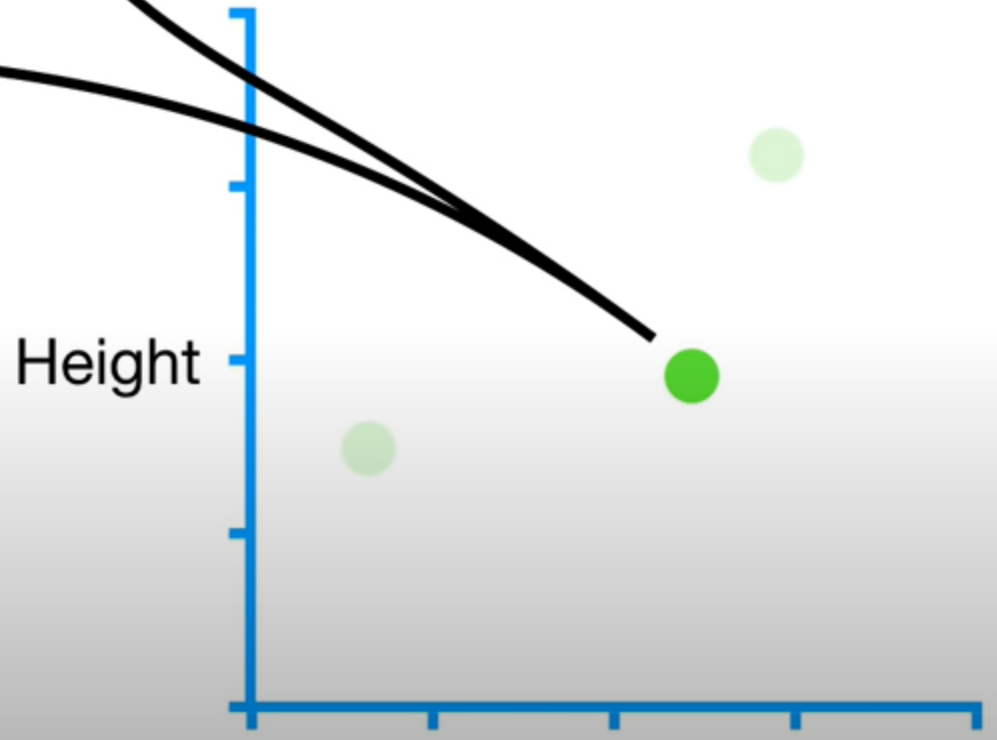
Stochastic Gradient Descent would randomly pick one sample for each step…



$$\frac{d}{d\ intercept}$$ Sum of squared residuals =

-2(**Height** - (intercept + slope × **Weight**)

…and just use that one sample to calculate the derivatives.



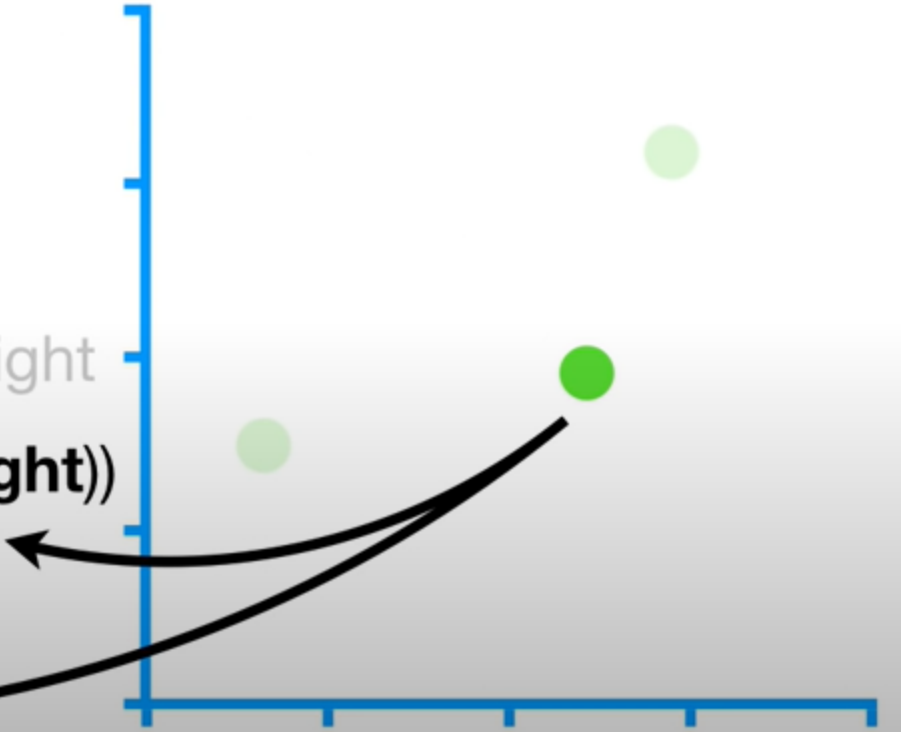
$$\frac{d}{d\ intercept}$$ Sum of squared residuals =

-2(**Height** - (intercept + slope × **Weight**)

…and just use that one sample to calculate the derivatives.

$$\frac{d}{d\ slope}$$ Sum of squared residuals =

-2 × **Weight**(**Height** - (intercept + slope × **Weight**))



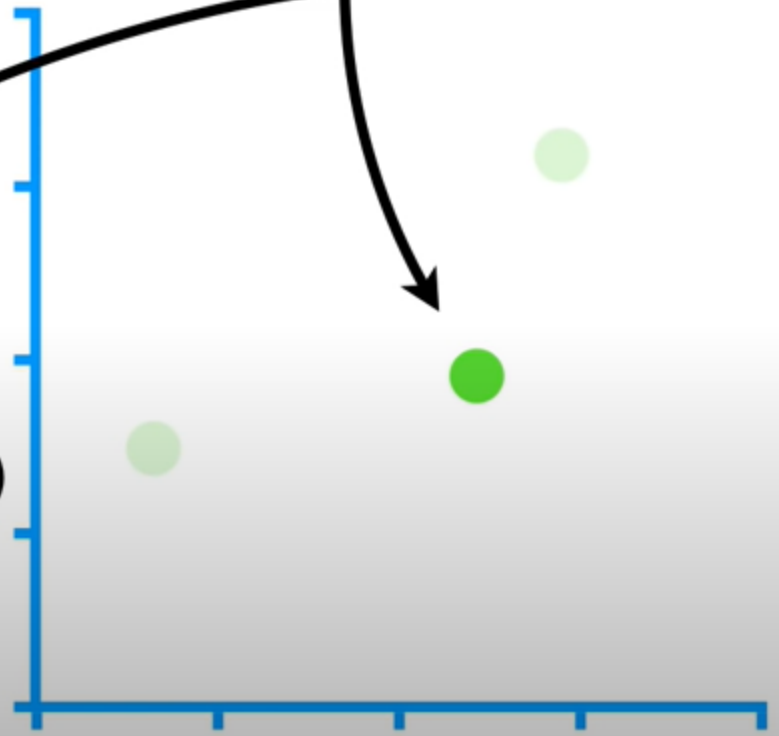
$\frac{d}{d\ intercept}$ Sum of squared residuals =

-2(**Height** - (intercept + slope × **Weight**)

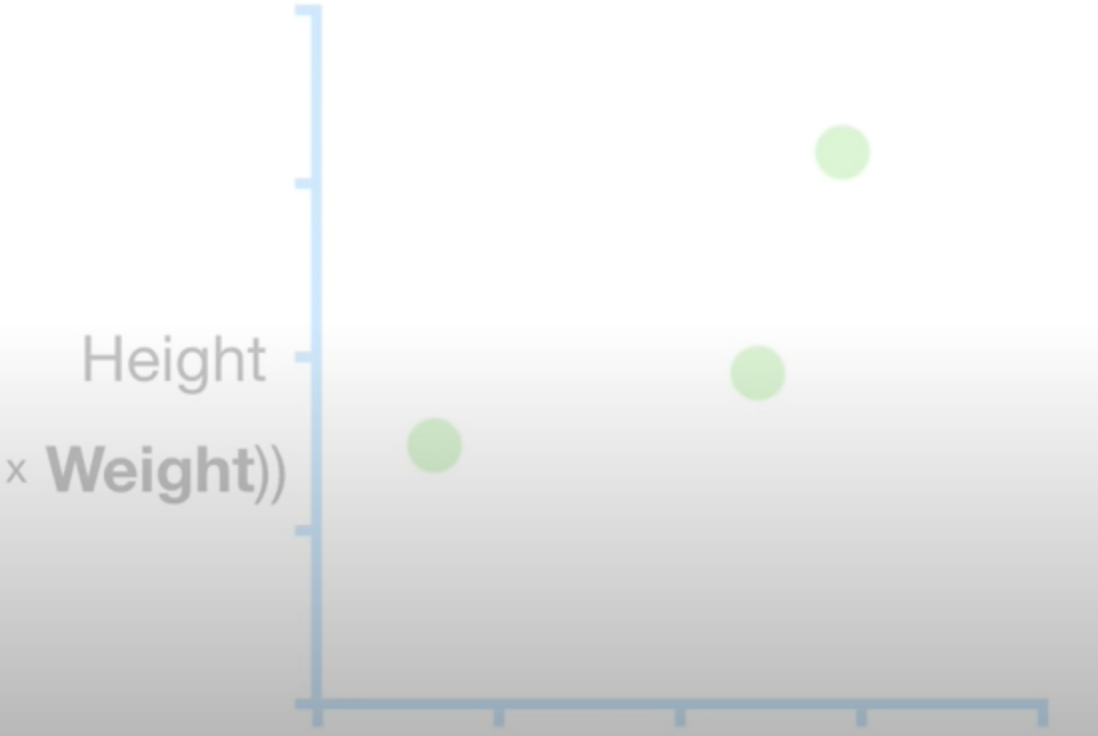
Thus, in this super simple example, **Stochastic Gradient Descent** reduced the number of terms computed by a factor of **3**.

$\frac{d}{d\ slope}$ Sum of squared residuals =

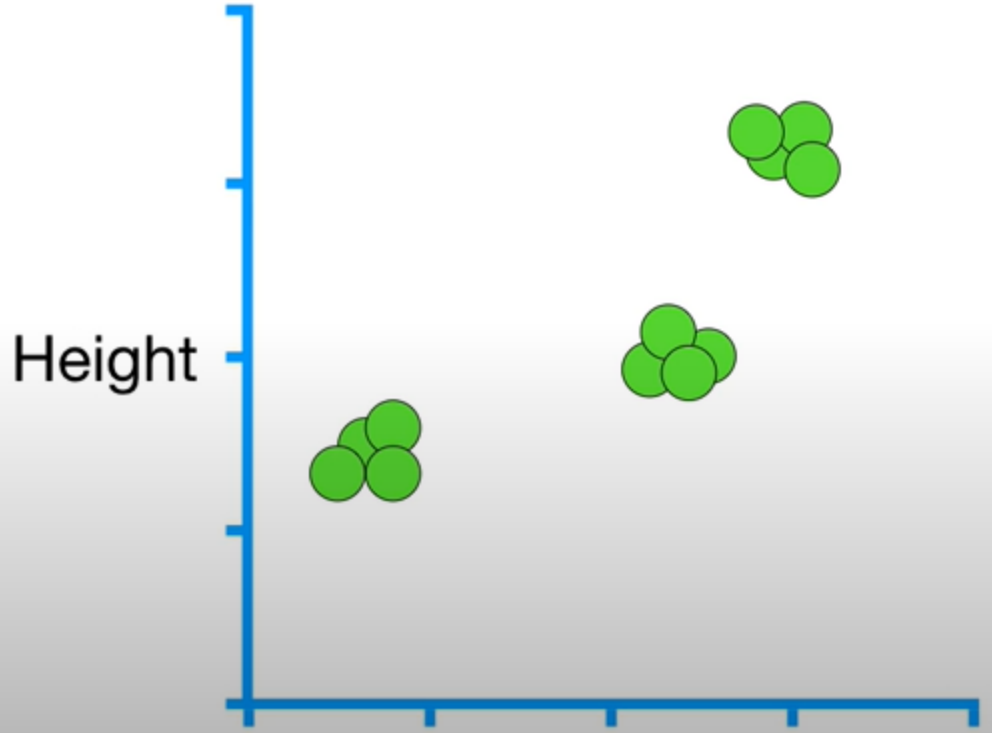
-2 × **Weight**(**Height** - (intercept + slope × **Weight**))



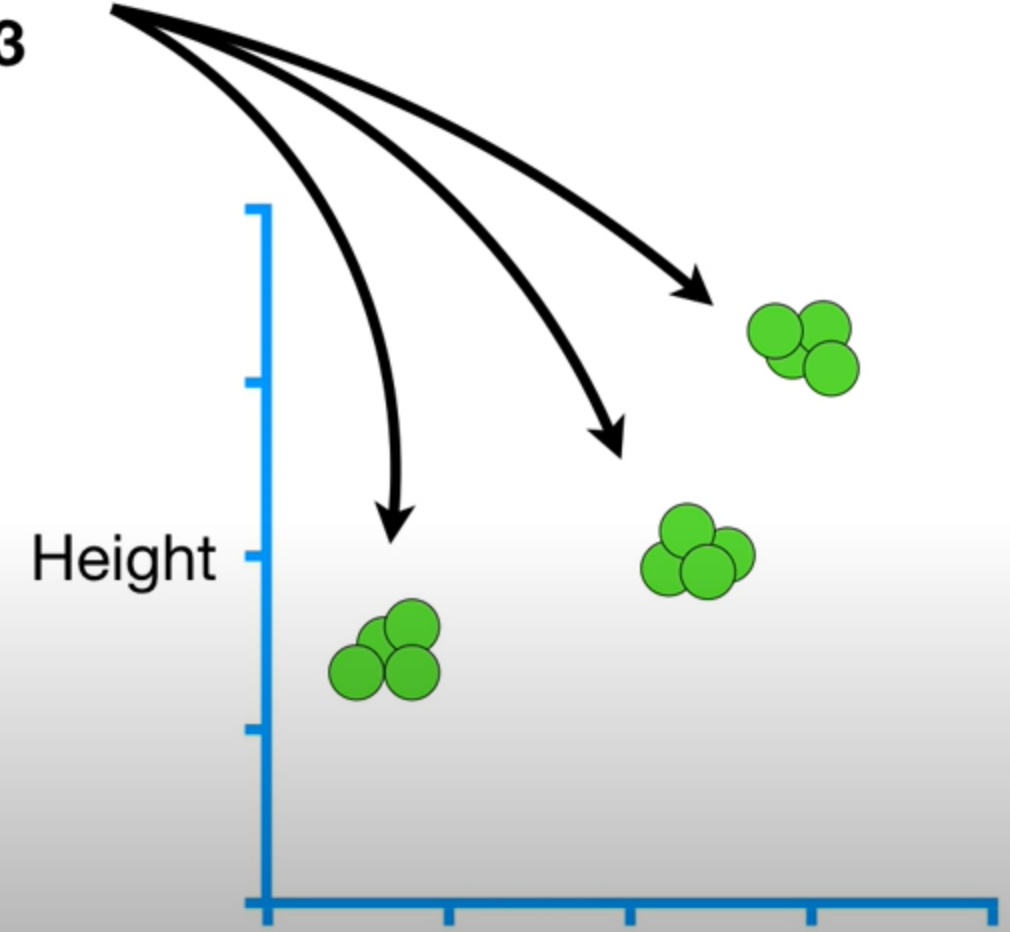
If we had **1,000,000** samples, then **Stochastic Gradient Descent** would reduce the amount terms computed by a factor of **1,000,000**.



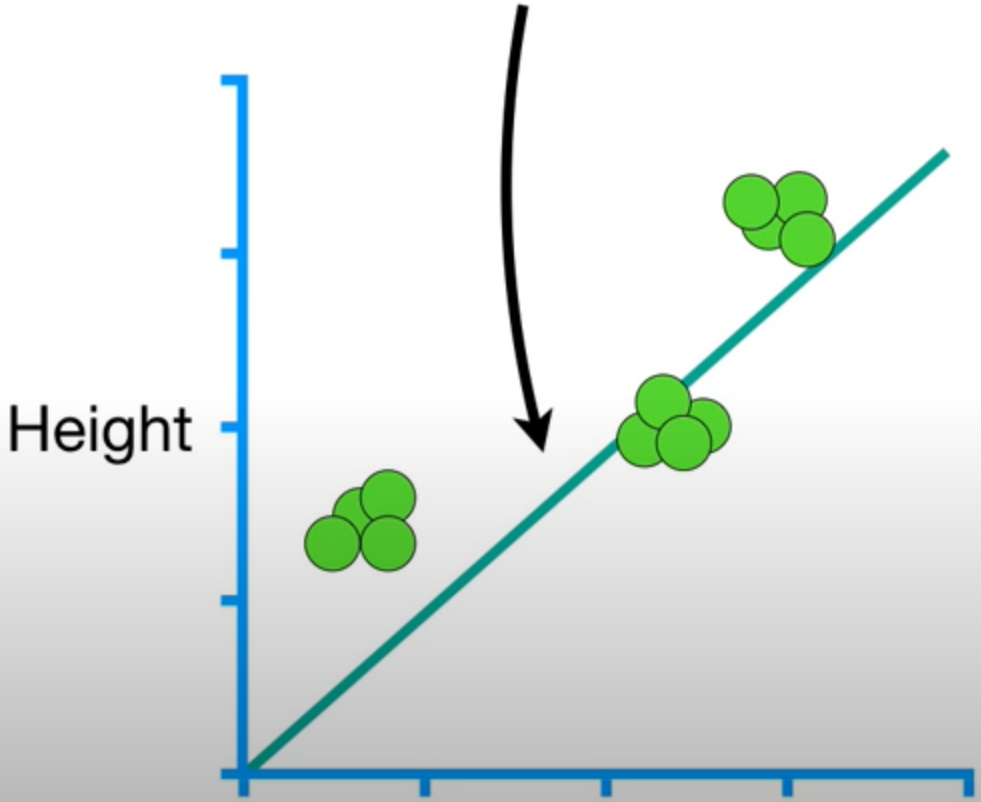
Stochastic Gradient Descent is especially useful when there are redundancies in the data.

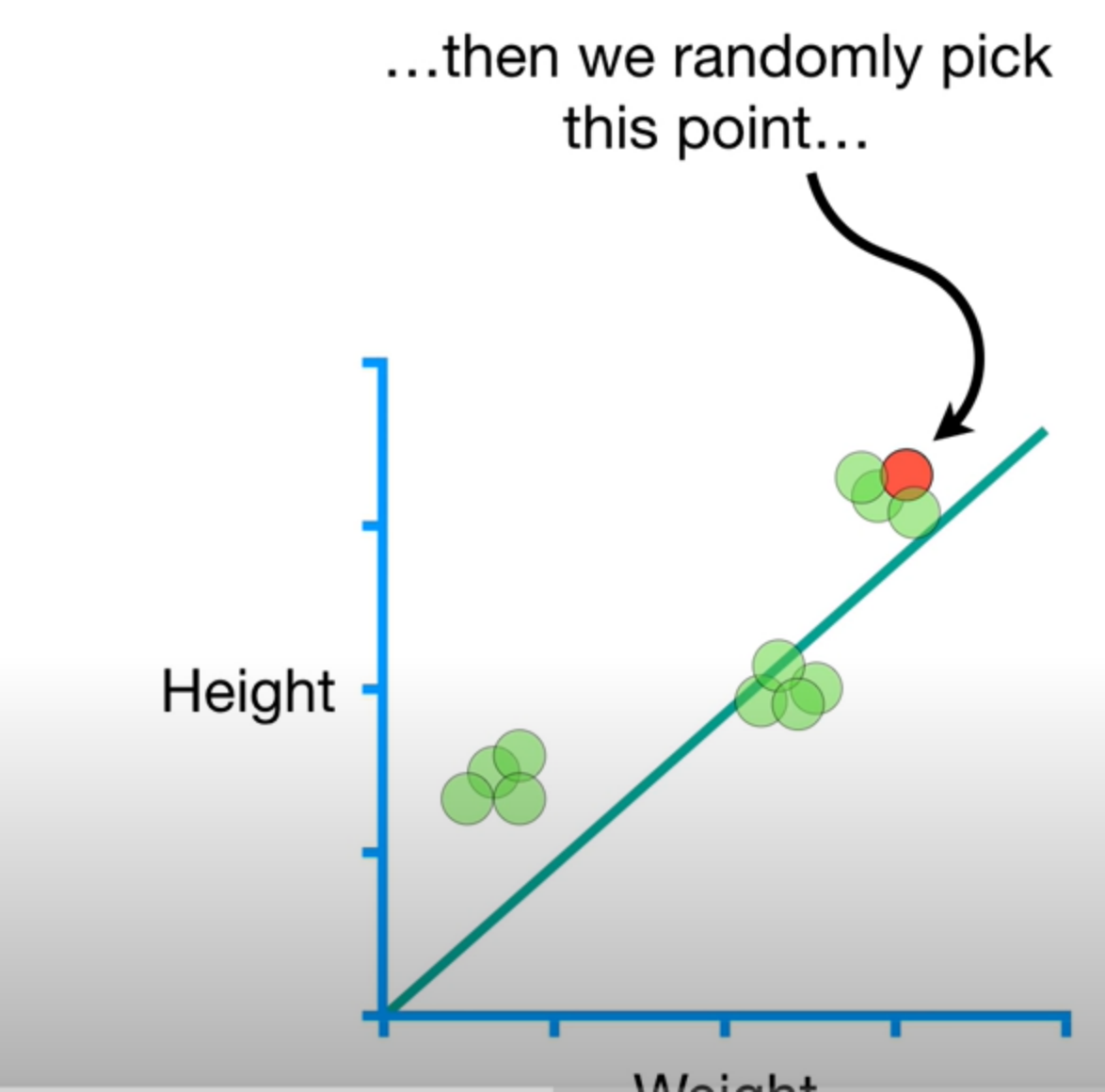


For example, we have **12** data points, but there is a lot of redundancy that forms **3** clusters.



So we start with a line with the **intercept** = **0** and the **slope** = **1**...




…then we randomly pick this point…
Height

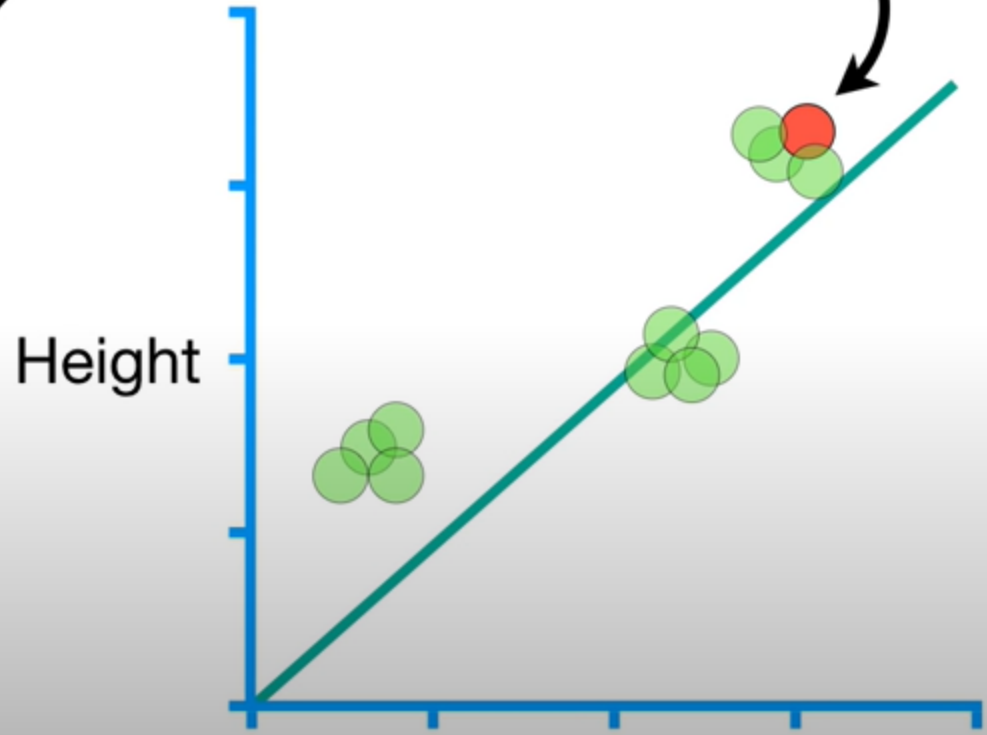
$$\frac{d}{d\ intercept}$$ Sum of squared residuals =

$$-2(\textbf{Height} - (0 + 1 \times \textbf{Weight})$$

…so we plug in the **Weight, 3**…

$$\frac{d}{d\ slope}$$ Sum of squared residuals =

$$-2 \times \textbf{Weight}(\textbf{Height} - (0 + 1 \times \textbf{Weight}))$$



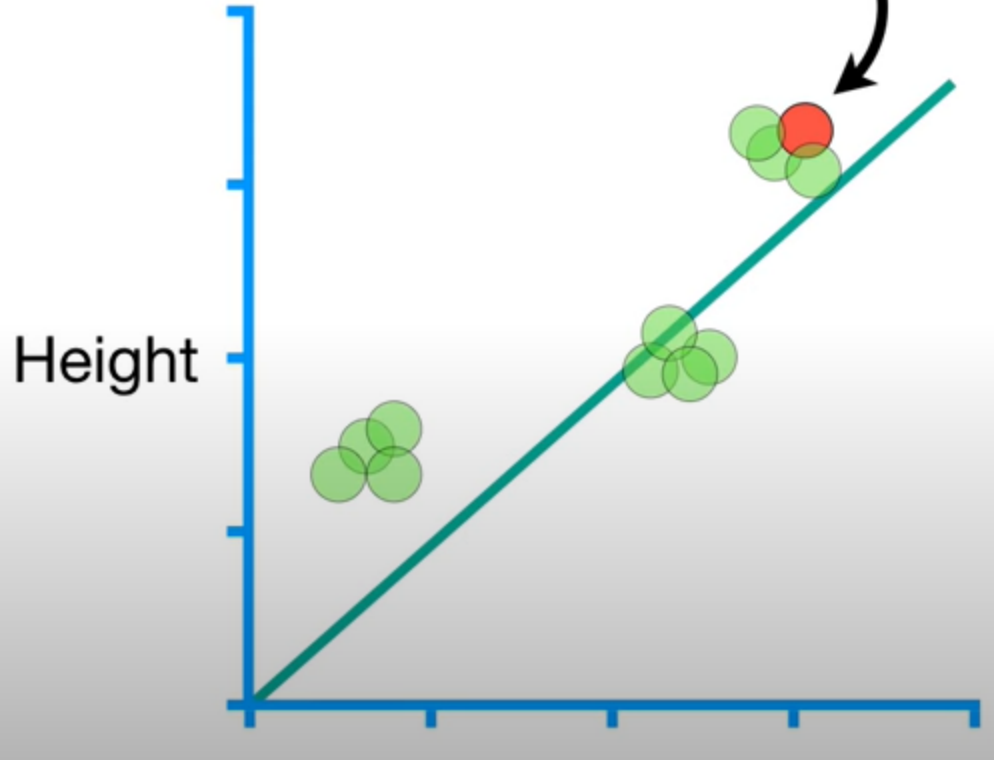
$\frac{d}{d\ intercept}$ Sum of squared residuals =

$-2(\mathbf{Height} - (0 + 1 \times \mathbf{3})$

…so we plug in the **Weight**, **3**…

$\frac{d}{d\ slope}$ Sum of squared residuals =

$-2 \times \mathbf{3}(\mathbf{Height} - (0 + 1 \times \mathbf{3}))$



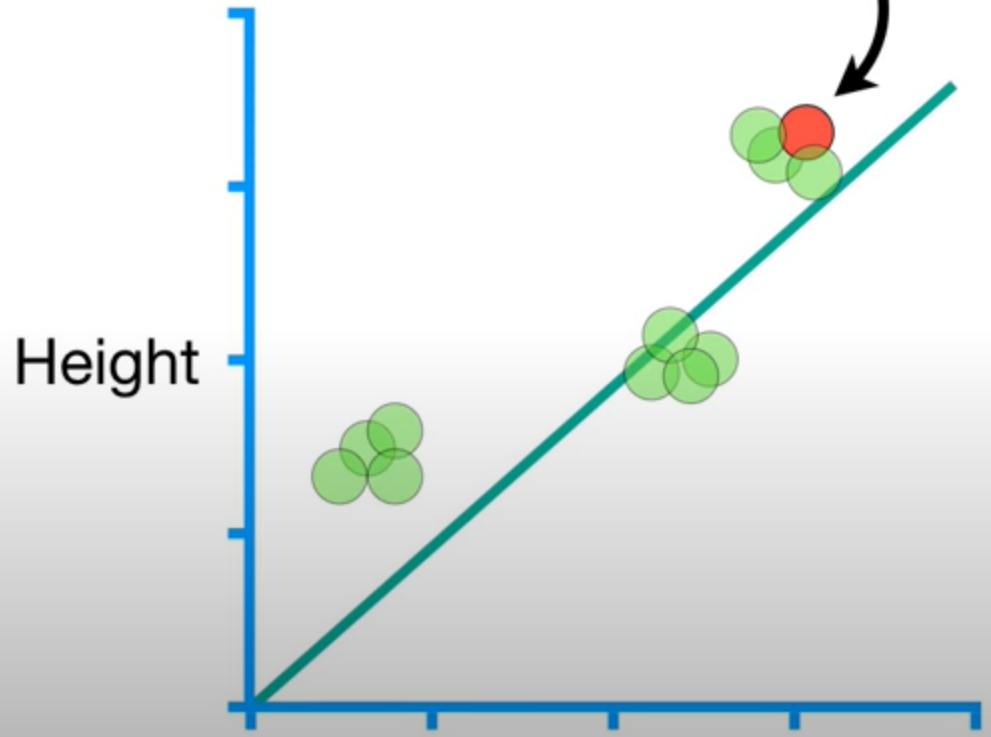
$$\frac{d}{d\,intercept}$$ Sum of squared residuals =

$$-2(\textbf{Height} - (0 + 1 \times \textbf{3})$$

…and **Height**, **3.3**…

$$\frac{d}{d\,slope}$$ Sum of squared residuals =

$$-2 \times \textbf{3}(\textbf{Height} - (0 + 1 \times \textbf{3}))$$



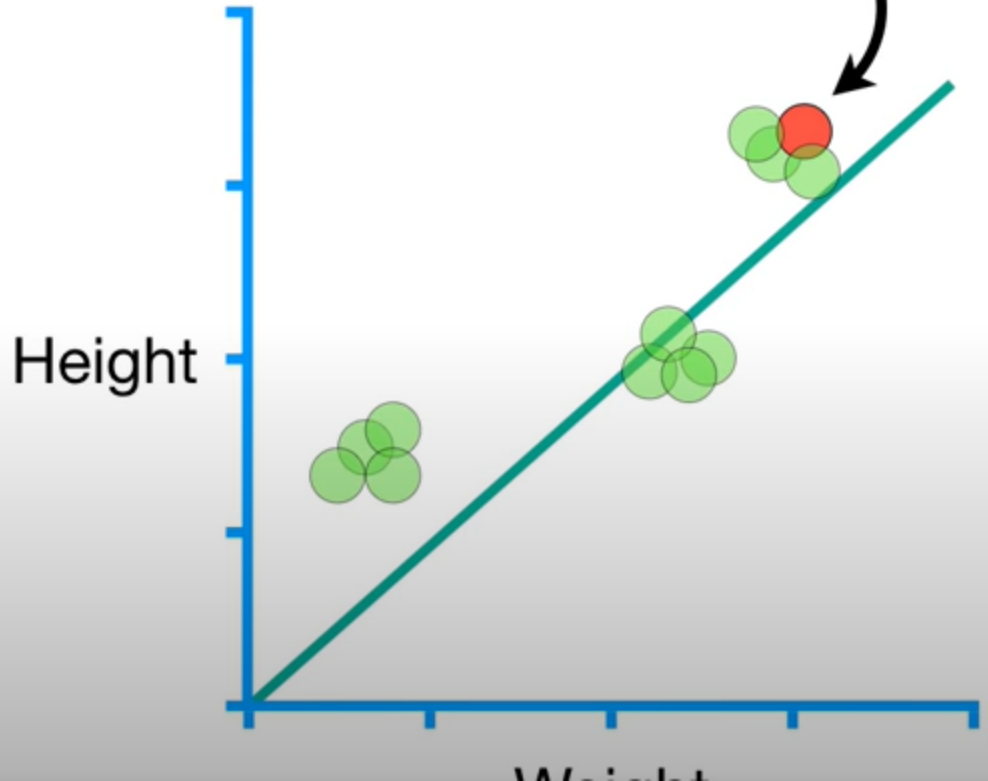
$$\frac{d}{d\ intercept}$$ Sum of squared residuals =

$$-2(\mathbf{3.3} - (0 + 1 \times \mathbf{3})$$

…and **Height**, **3.3**…

$$\frac{d}{d\ slope}$$ Sum of squared residuals =

$$-2 \times \mathbf{3}(\mathbf{3.3} - (0 + 1 \times \mathbf{3}))$$



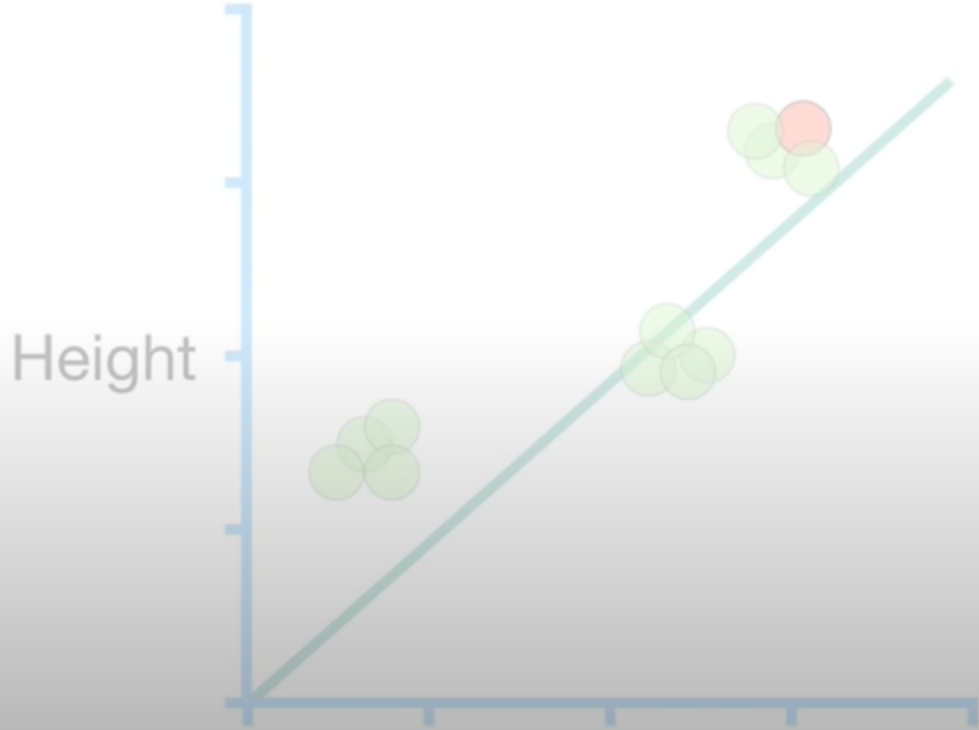
$$\frac{d}{d\ intercept}$$ Sum of squared residuals =

$$-2(\mathbf{3.3} - (0 + 1 \times \mathbf{3}) = -0.6$$

…do the math…

$$\frac{d}{d\ slope}$$ Sum of squared residuals =

$$-2 \times \mathbf{3}(\mathbf{3.3} - (0 + 1 \times \mathbf{3})) = -1.8$$



$\frac{d}{d\ intercept}$ Sum of squared residuals =

$-2(\mathbf{3.3} - (0 + 1 \times \mathbf{3}) = -0.6$

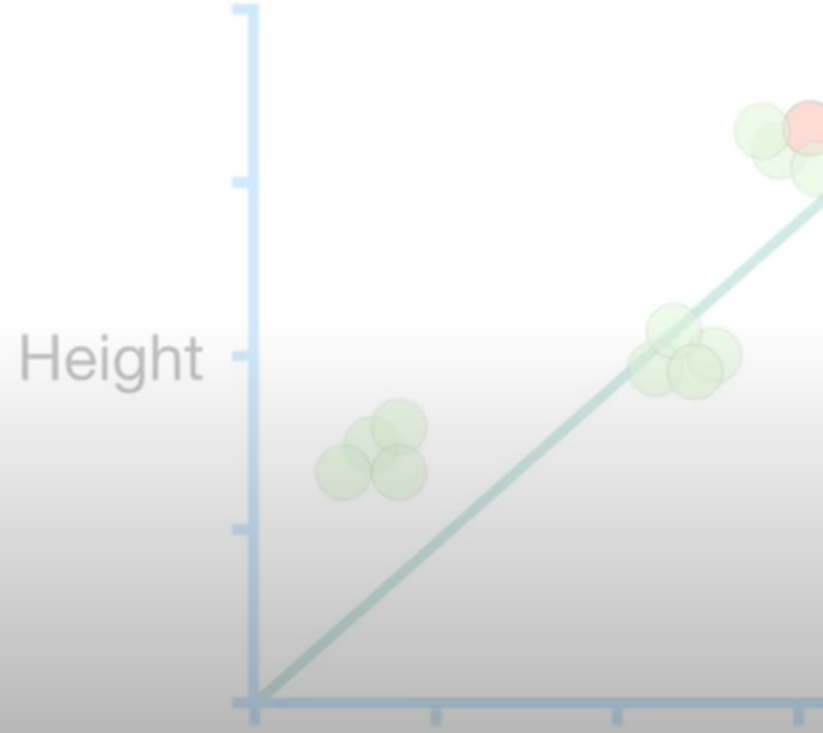
…plug in the slopes…

Step Size$_{\text{Intercept}}$ = **Slope** x **Learning Rate**

Step Size$_{\text{Slope}}$ = **Slope** x **Learning Rate**

$\frac{d}{d\ slope}$ Sum of squared residuals =

$-2 \times \mathbf{3}(\mathbf{3.3} - (0 + 1 \times \mathbf{3})) = -1.8$



$\frac{d}{d\ intercept}$ Sum of squared residuals =

$-2(\mathbf{3.3} - (0 + 1 \times \mathbf{3}) = -0.6$

Step Size$_{\text{Intercept}}$ = -0.6 x **Learning Rate**

Step Size$_{\text{Slope}}$ = -1.8 x **Learning Rate**

$\frac{d}{d\ slope}$ Sum of squared residuals =

$-2 \times \mathbf{3}(\mathbf{3.3} - (0 + 1 \times \mathbf{3})) = -1.8$

$\frac{d}{d\,intercept}$ Sum of squared residuals =

$-2(\mathbf{3.3} - (0 + 1 \times \mathbf{3}) = -0.6$

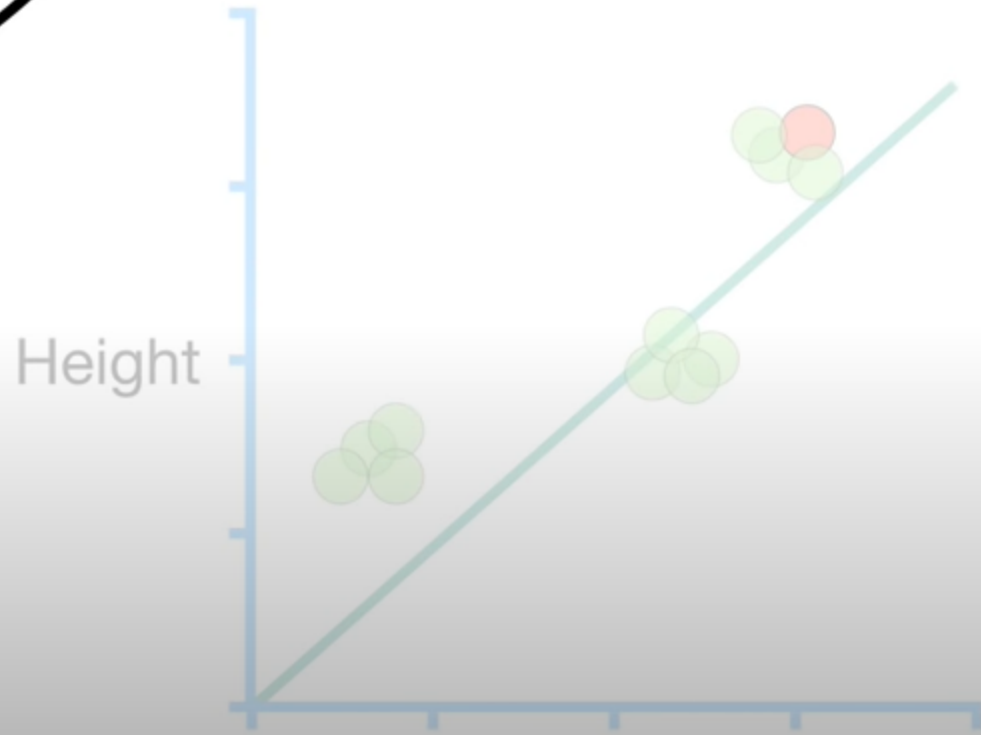
…then multiply by the **Learning Rate**.

Step Size$_{\text{Intercept}}$ = -0.6 × **Learning Rate**

Step Size$_{\text{Slope}}$ = -1.8 × **Learning Rate**

$\frac{d}{d\,slope}$ Sum of squared residuals =

$-2 \times \mathbf{3}(\mathbf{3.3} - (0 + 1 \times \mathbf{3})) = -1.8$



$\frac{d}{d\ intercept}$ Sum of squared residuals =

$-2(\mathbf{3.3} - (0 + 1 \times \mathbf{3}) = -0.6$

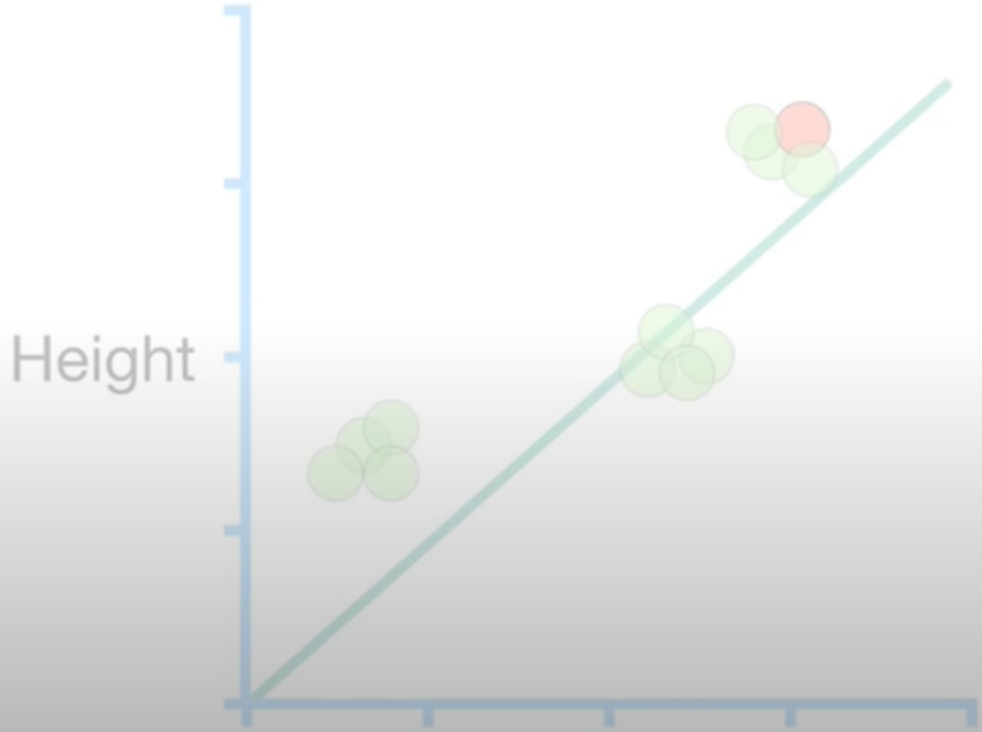
Step Size$_{\text{Intercept}}$ = -0.6 × **Learning Rate**

Step Size$_{\text{Slope}}$ = -1.8 × **Learning Rate**

$\frac{d}{d\ slope}$ Sum of squared residuals =

$-2 \times \mathbf{3}(\mathbf{3.3} - (0 + 1 \times \mathbf{3})) = -1.8$

NOTE: Just like with regular **Gradient Descent**, **Stochastic Gradient Descent** is sensitive to the value you choose for the **Learning Rate**…



$\frac{d}{d\,intercept}$ Sum of squared residuals =

$-2(\mathbf{3.3} - (0 + 1 \times \mathbf{3}) = -0.6$

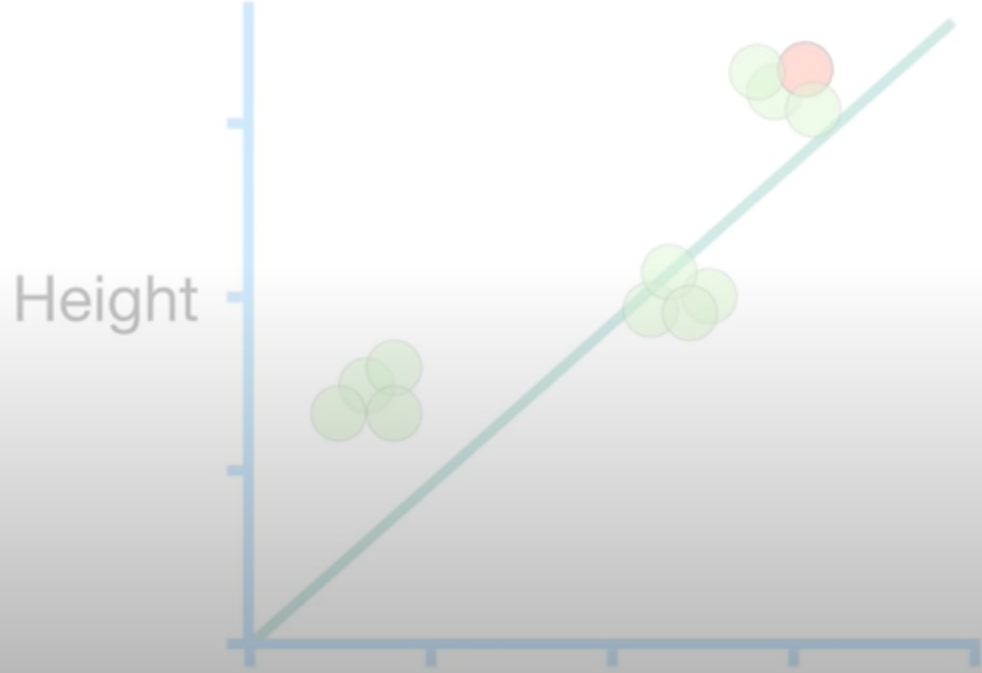
Step Size$_{Intercept}$ = -0.6 × **Learning Rate**

Step Size$_{Slope}$ = -1.8 × **Learning Rate**

$\frac{d}{d\,slope}$ Sum of squared residuals =

$-2 \times \mathbf{3}(\mathbf{3.3} - (0 + 1 \times \mathbf{3})) = -1.8$

…and just like for regular **Gradient Descent**, the general strategy is to start with a relatively *large* **Learning Rate** and make it *smaller* with each step…



$\frac{d}{d\,intercept}$ Sum of squared residuals =

$-2(\mathbf{3.3} - (0 + 1 \times \mathbf{3}) = -0.6$

…and lastly, just like for regular **Gradient Descent**, many implementations of **Stochastic Gradient Descent** will take care of this for you by default.

Step Size$_{Intercept}$ = -0.6 × **Learning Rate**

Step Size$_{Slope}$ = -1.8 × **Learning Rate**

$\frac{d}{d\,slope}$ Sum of squared residuals =

Height

$\frac{d}{d\ intercept}$ Sum of squared residuals =

$-2(\mathbf{3.3} - (0 + 1 \times \mathbf{3}) = -0.6$

TERMINOLOGY ALERT!!!

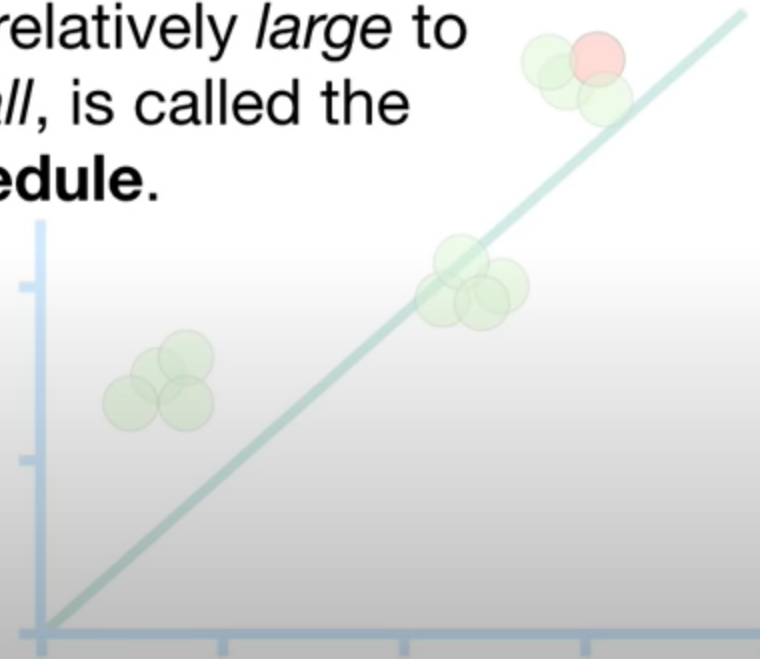
Step Size$_{\text{Intercept}}$ = -0.6 × **Learning Rate**

Step Size$_{\text{Slope}}$ = -1.8 × **Learning Rate**

The way the **Learning Rate** changes, from relatively *large* to relatively *small*, is called the **schedule**.

$\frac{d}{d\ slope}$ Sum of squared residuals =

$-2 \times \mathbf{3}(\mathbf{3.3} - (0 + 1 \times \mathbf{3})) = -1.8$



$\frac{d}{d\ intercept}$ Sum of squared residuals =

$-2(\mathbf{3.3} - (0 + 1 \times \mathbf{3}) = -0.6$

Step Size$_{\text{Intercept}}$ = -0.6 × **Learning Rate**

Step Size$_{\text{Slope}}$ = -1.8 × **Learning Rate**

$\frac{d}{d\ slope}$ Sum of squared residuals =

$-2 \times \mathbf{3}(\mathbf{3.3} - (0 + 1 \times \mathbf{3})) = -1.8$

TERMINOLOGY ALERT!!!

The way the **Learning Rate** changes, from relatively *large* to relatively *small*, is called the **schedule**.

So if you fail to converge on parameter estimates, try futzing with this setting.

$\frac{d}{d\,intercept}$ Sum of squared residuals =

$-2(\mathbf{3.3} - (0 + 1 \times \mathbf{3}) = -0.6$

In this simple example, however, we'll just setting the **Learning Rate** to **0.01**.

Step Size$_{Intercept}$ = -0.6 × **Learning Rate**

Step Size$_{Slope}$ = -1.8 × **Learning Rate**

$\frac{d}{d\,slope}$ Sum of squared residuals =

Height

$\frac{d}{d\ intercept}$ Sum of squared residuals =

$-2(\mathbf{3.3} - (0 + 1 \times \mathbf{3}) = -0.6$

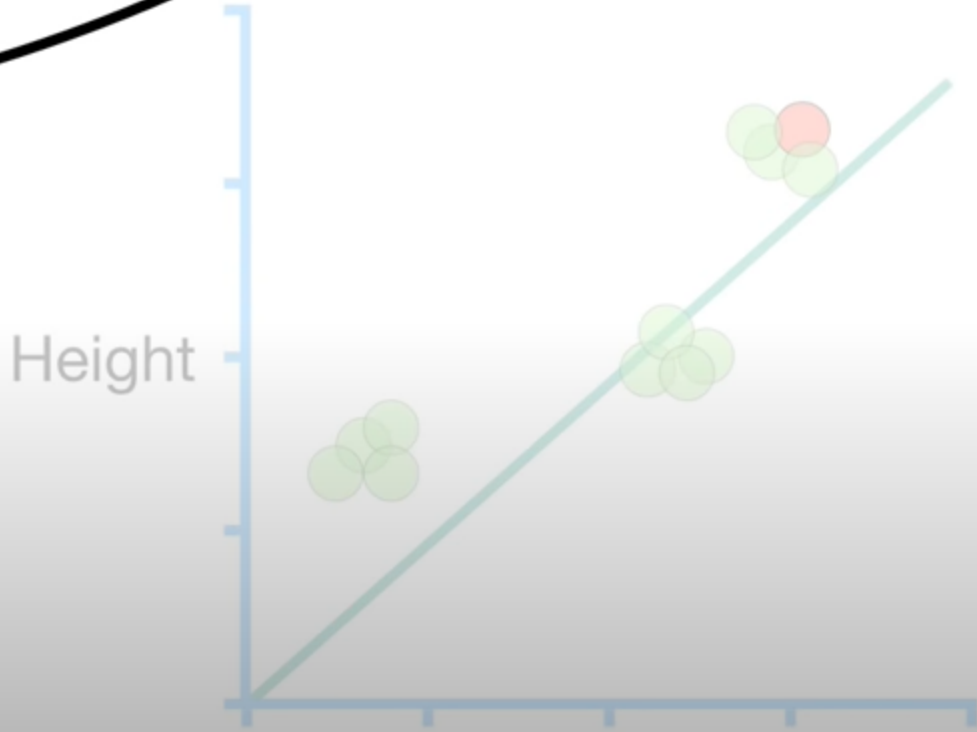
In this simple example, however, we'll just setting the **Learning Rate** to **0.01**.

Step Size$_{\text{Intercept}}$ = -0.6 × 0.01

Step Size$_{\text{Slope}}$ = -1.8 × 0.01

$\frac{d}{d\ slope}$ Sum of squared residuals =

$-2 \times \mathbf{3}(\mathbf{3.3} - (0 + 1 \times \mathbf{3})) = -1.8$



$\frac{d}{d\ intercept}$ Sum of squared residuals =

$-2(\mathbf{3.3} - (0 + 1 \times \mathbf{3}) = -0.6$

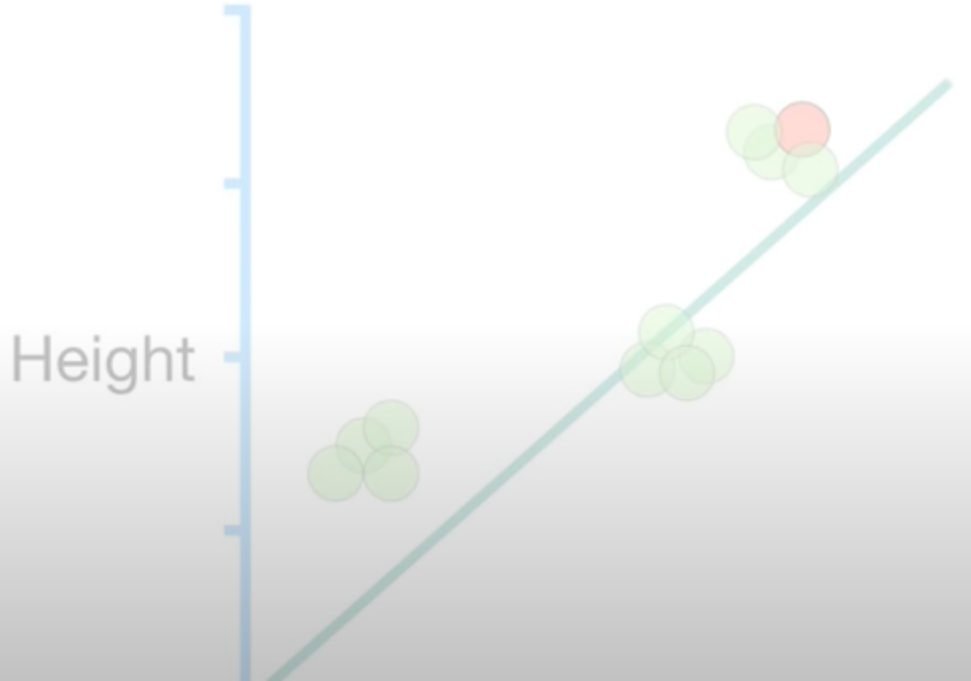
Now we do the math…

Step Size$_{\text{Intercept}}$ = -0.6 x 0.01 = **-0.006**

Step Size$_{\text{Slope}}$ = -1.8 x 0.01 = **-0.018**

$\frac{d}{d\ slope}$ Sum of squared residuals =

$-2 \times \mathbf{3}(\mathbf{3.3} - (0 + 1 \times \mathbf{3})) = -1.8$

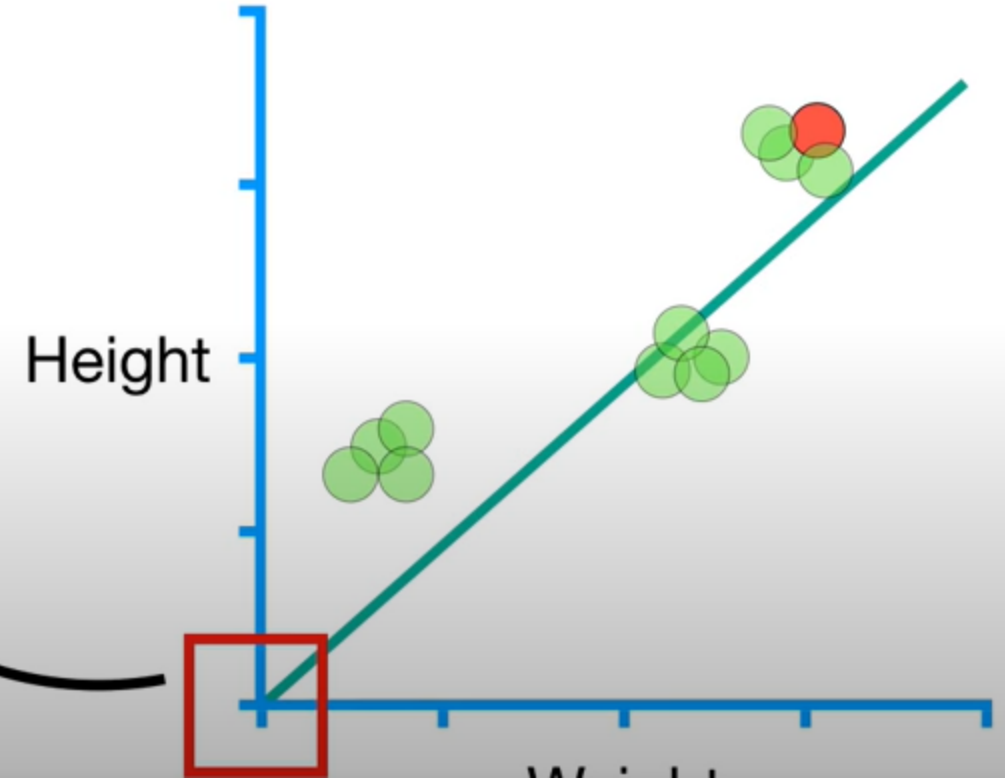


New Intercept = **Old Intercept** - **Step Size**

…calculate the new **intercept**…

Step Size$_{\text{Intercept}}$ = -0.6 x 0.01 = **-0.006**

Step Size$_{\text{Slope}}$ = -1.8 x 0.01 = **-0.018**

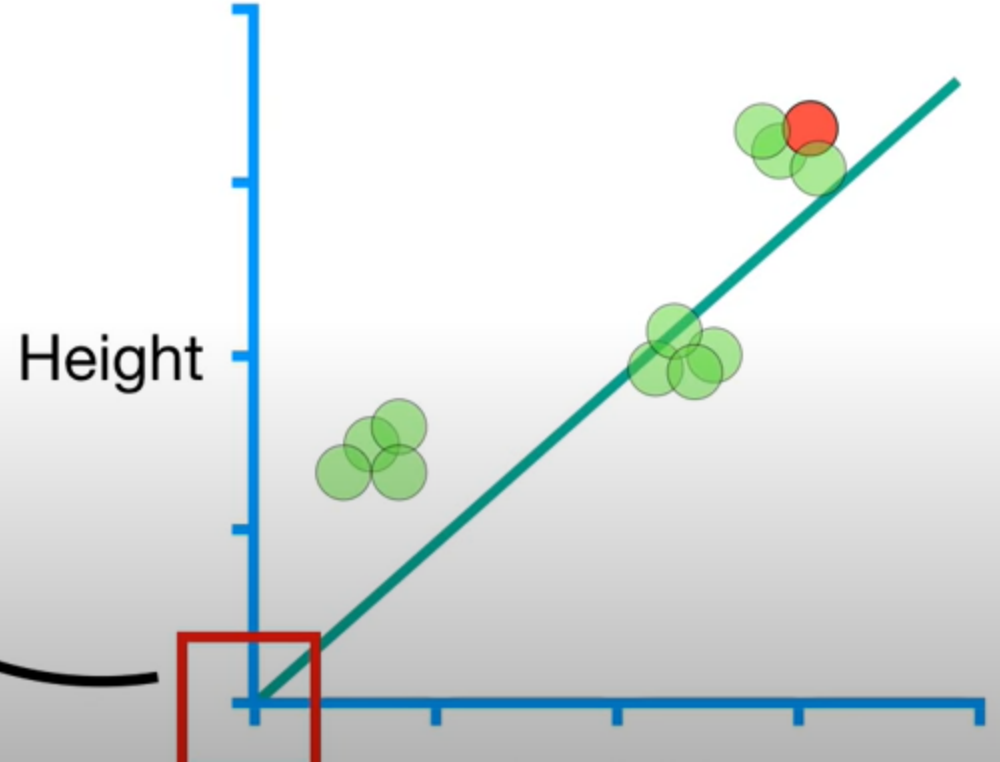


New Intercept = 0 - **Step Size**

Step Size$_{\text{Intercept}}$ = -0.6 × 0.01 = **-0.006**

Step Size$_{\text{Slope}}$ = -1.8 × 0.01 = **-0.018**

…calculate the new **intercept**…

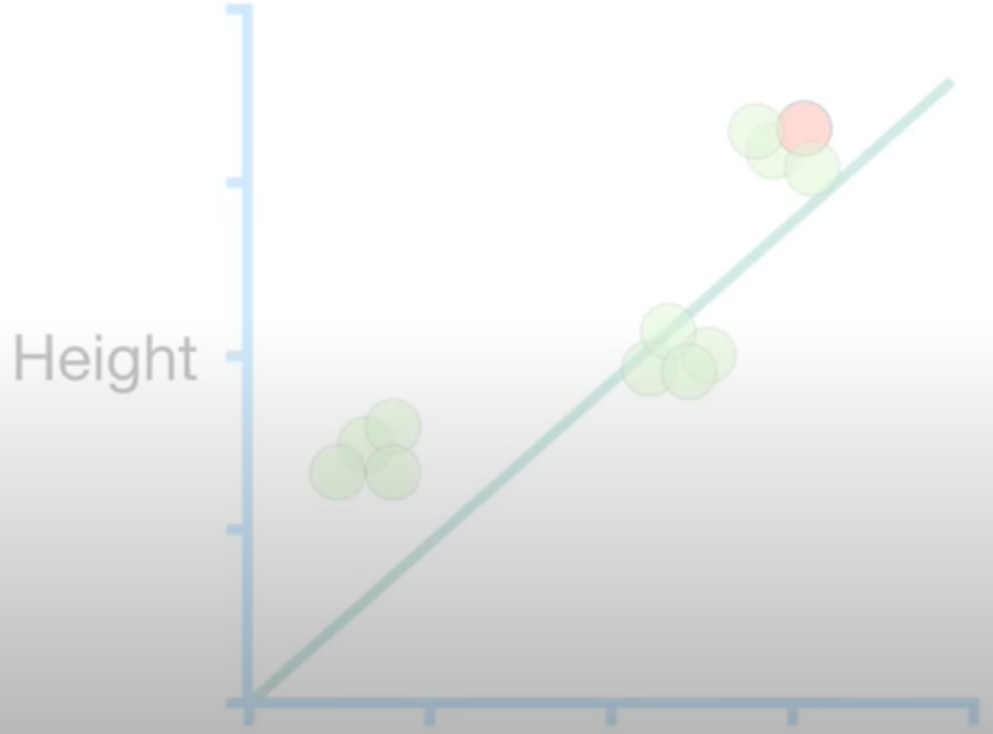


New Intercept = 0 - -0.006

Step Size$_{\text{Intercept}}$ = -0.6 x 0.01 = **-0.006**

Step Size$_{\text{Slope}}$ = -1.8 x 0.01 = **-0.018**

…calculate the new **intercept**…

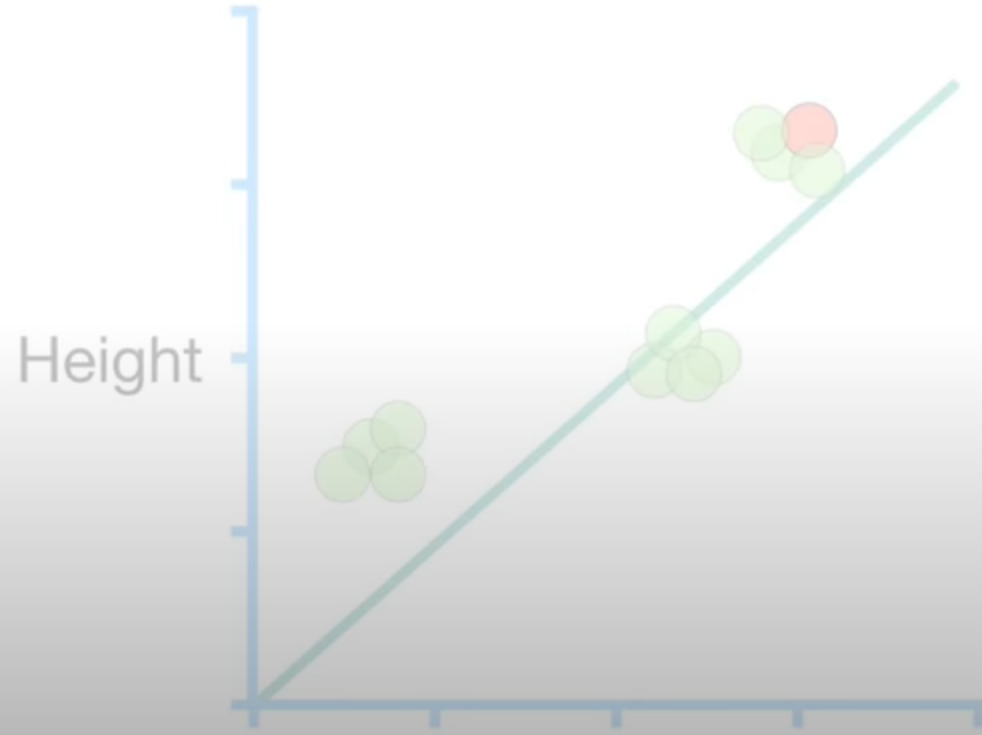


New Intercept = 0 - -0.006 = 0.006

Step Size$_{\text{Intercept}}$ = -0.6 x 0.01 = **-0.006**

Step Size$_{\text{Slope}}$ = -1.8 x 0.01 = **-0.018**

…calculate the new **intercept**…



New Intercept = 0 - -0.006 = 0.006

…and the new **slope**.

Step Size$_{\text{Intercept}}$ = -0.6 × 0.01 = **-0.006**

Step Size$_{\text{Slope}}$ = -1.8 × 0.01 = **-0.018**

New Slope = **Old Slope** - **Step Size**



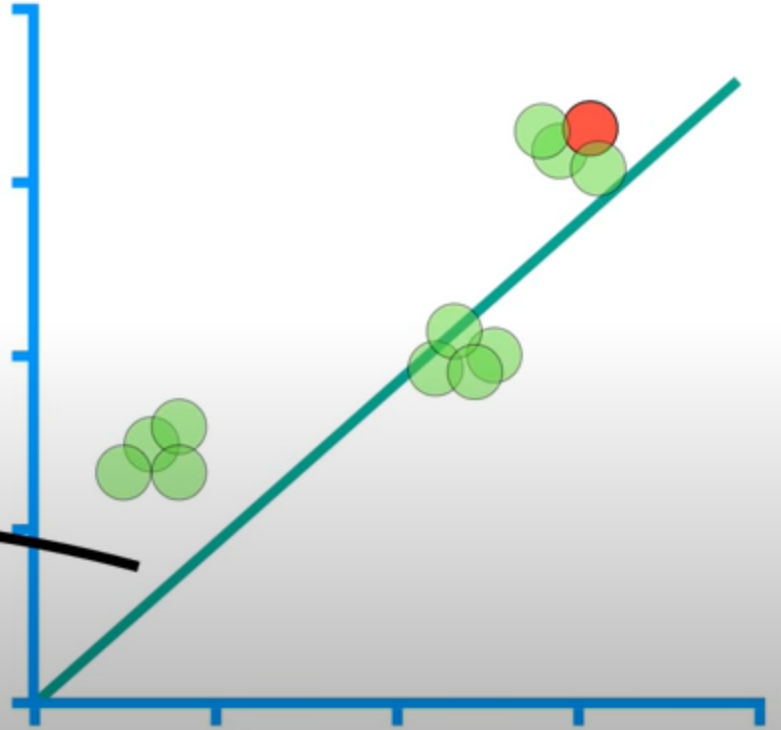
New Intercept = 0 - -0.006 = 0.006

Step Size$_{\text{Intercept}}$ = -0.6 x 0.01 = **-0.006**

…and the new **slope**.

Step Size$_{\text{Slope}}$ = -1.8 x 0.01 = **-0.018**

New Slope = 1 - **Step Size**



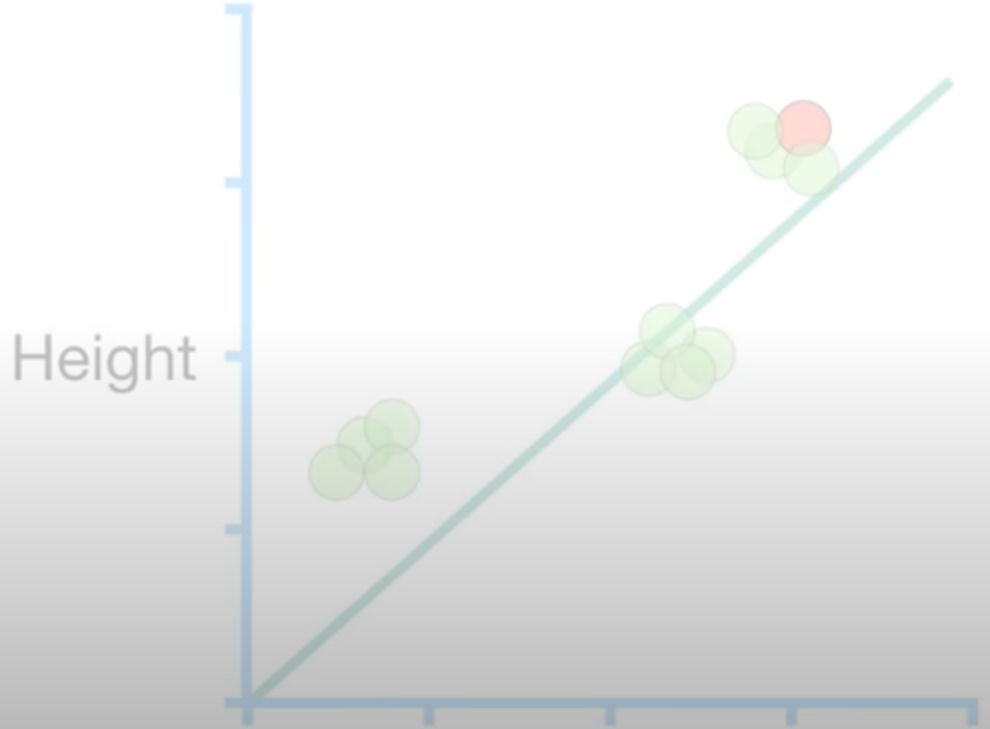
New Intercept = 0 - -0.006 = 0.006

Step Size$_{\text{Intercept}}$ = -0.6 x 0.01 = **-0.006**

Step Size$_{\text{Slope}}$ = -1.8 x 0.01 = **-0.018**

New Slope = 1 - -0.018

…and the new **slope**.



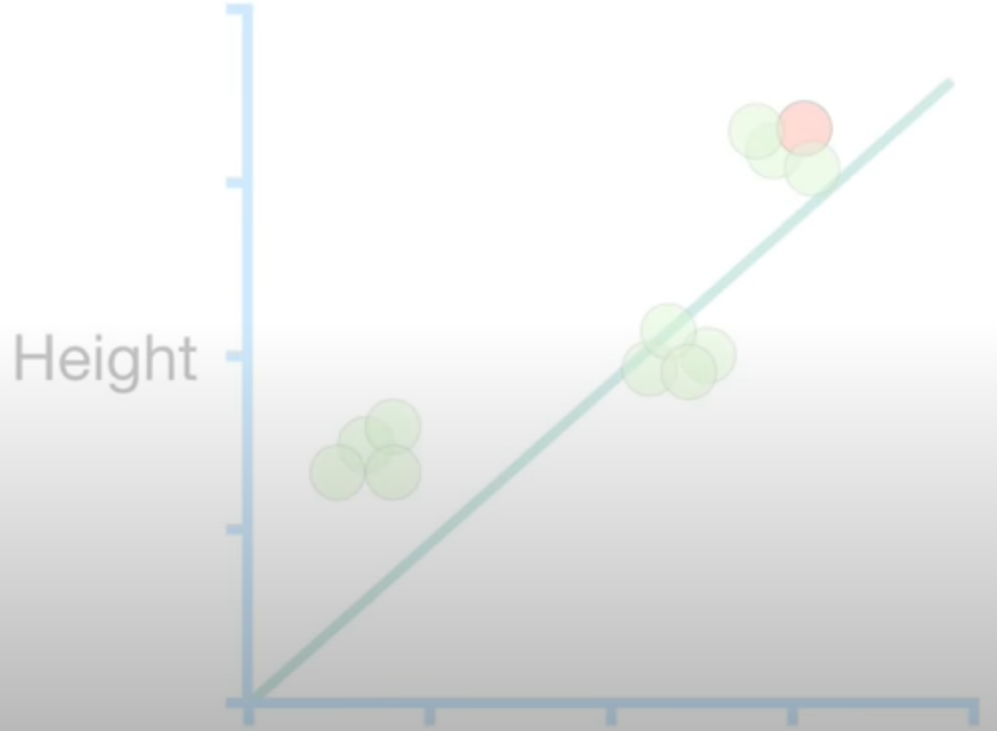
New Intercept = 0 - -0.006 = 0.006

Step Size$_{\text{Intercept}}$ = -0.6 x 0.01 = **-0.006**

Step Size$_{\text{Slope}}$ = -1.8 x 0.01 = **-0.018**

New Slope = 1 - -0.018 = 1.018

…and the new **slope**.

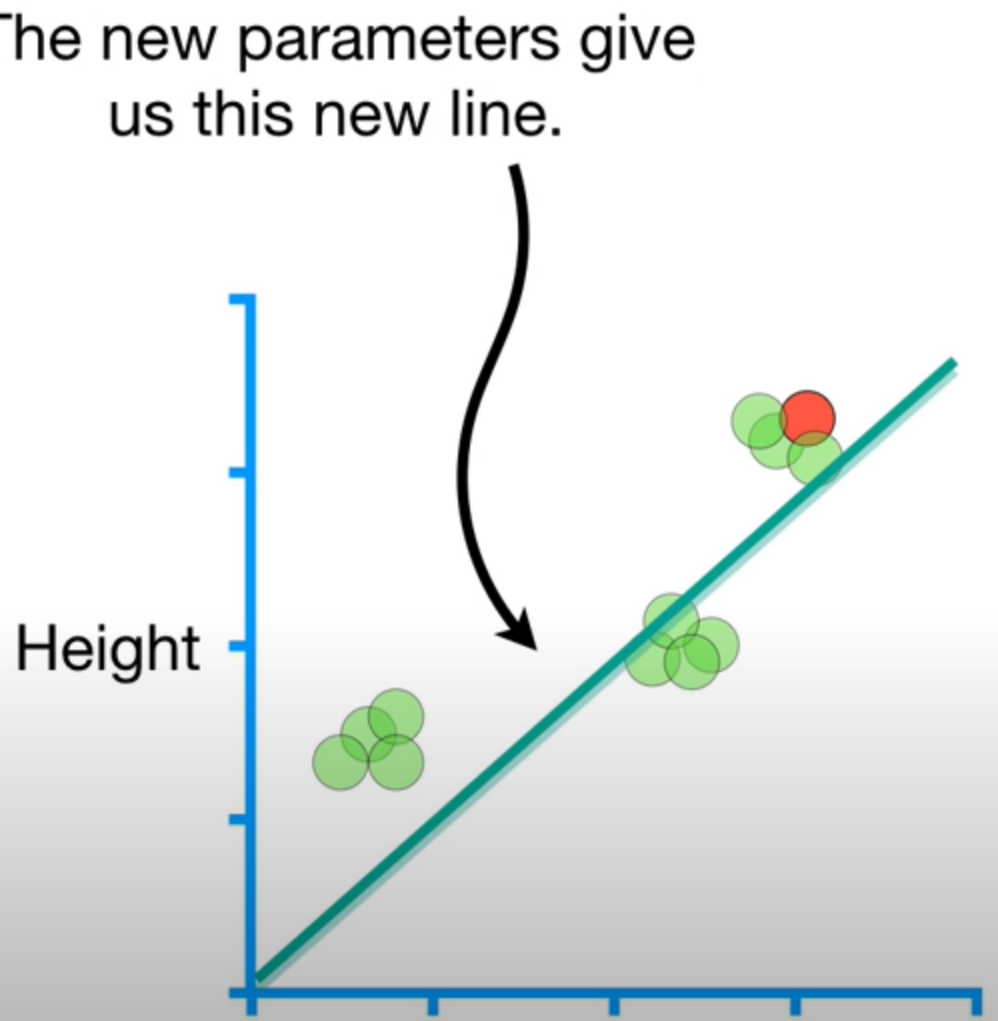


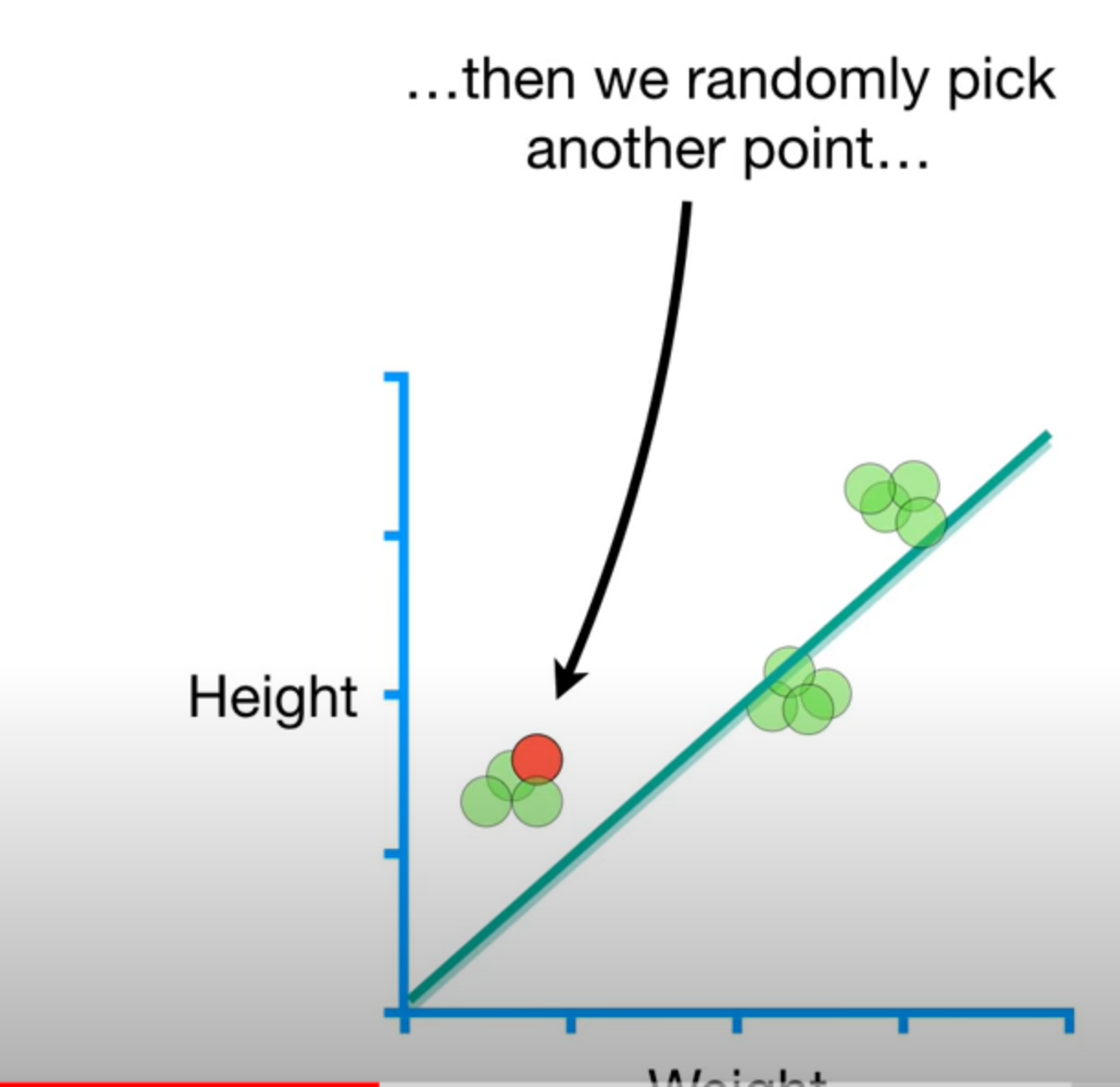
New Intercept = 0 - -0.006 = 0.006

New Slope = 1 - -0.018 = 1.018

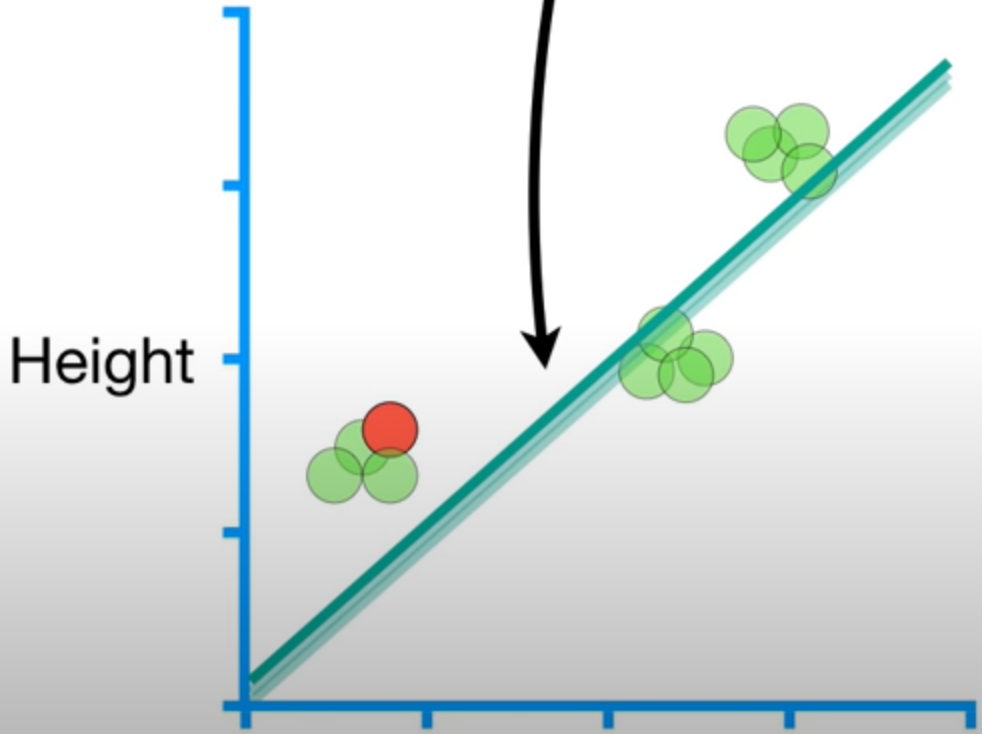
New Intercept = 0 - -0.006 = 0.006

New Slope = 1 - -0.018 = 1.018

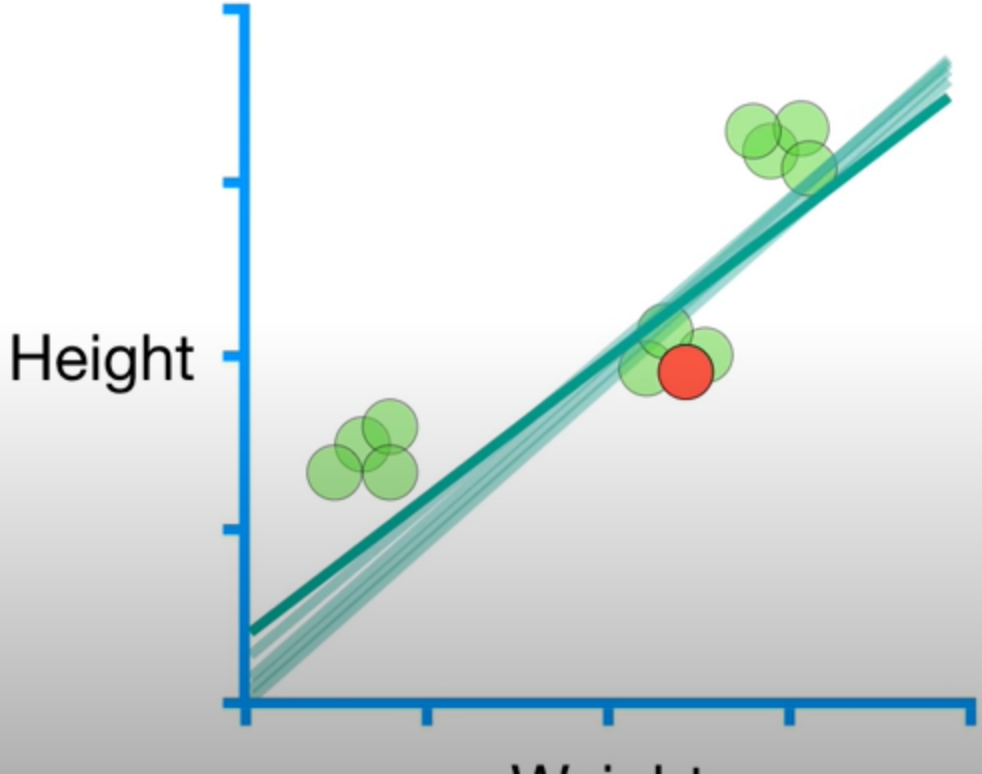



…then we randomly pick another point…
Height

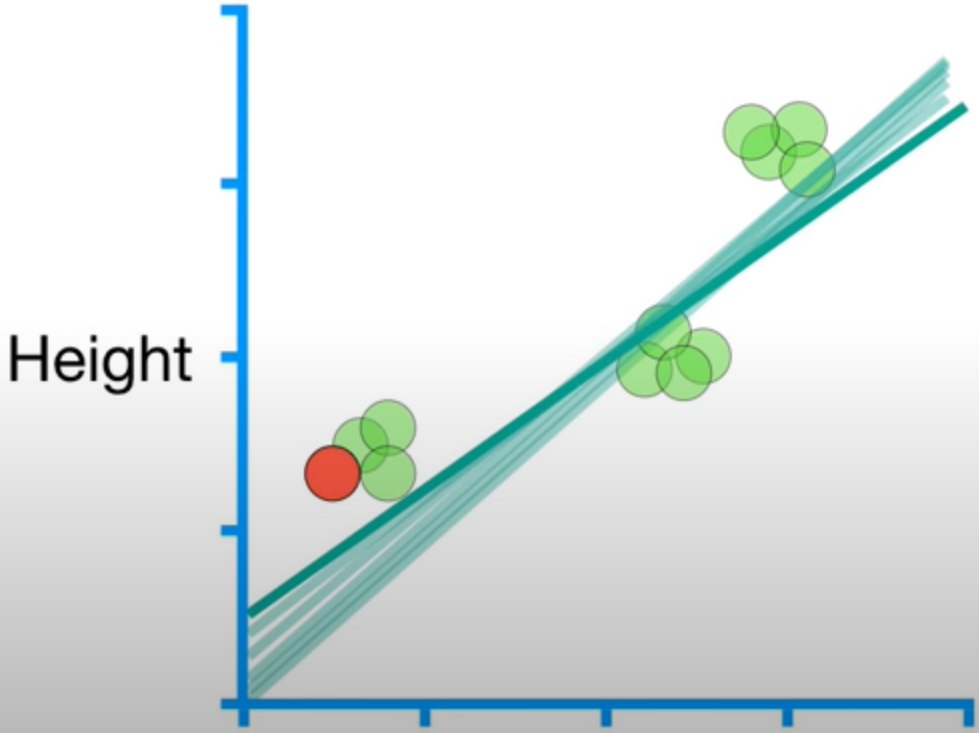
…and calculate the **intercept** and **slope** for another line.



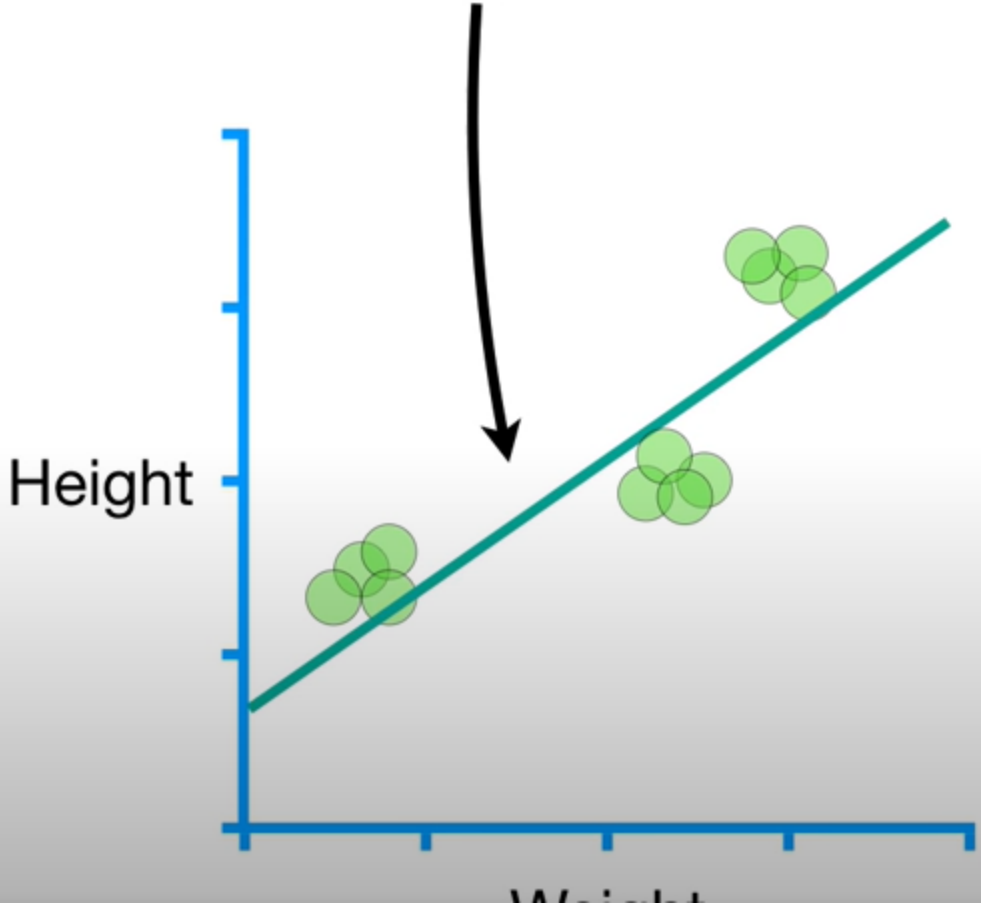
Then we just repeat everything a bunch of times…



Then we just repeat everything a bunch of times…

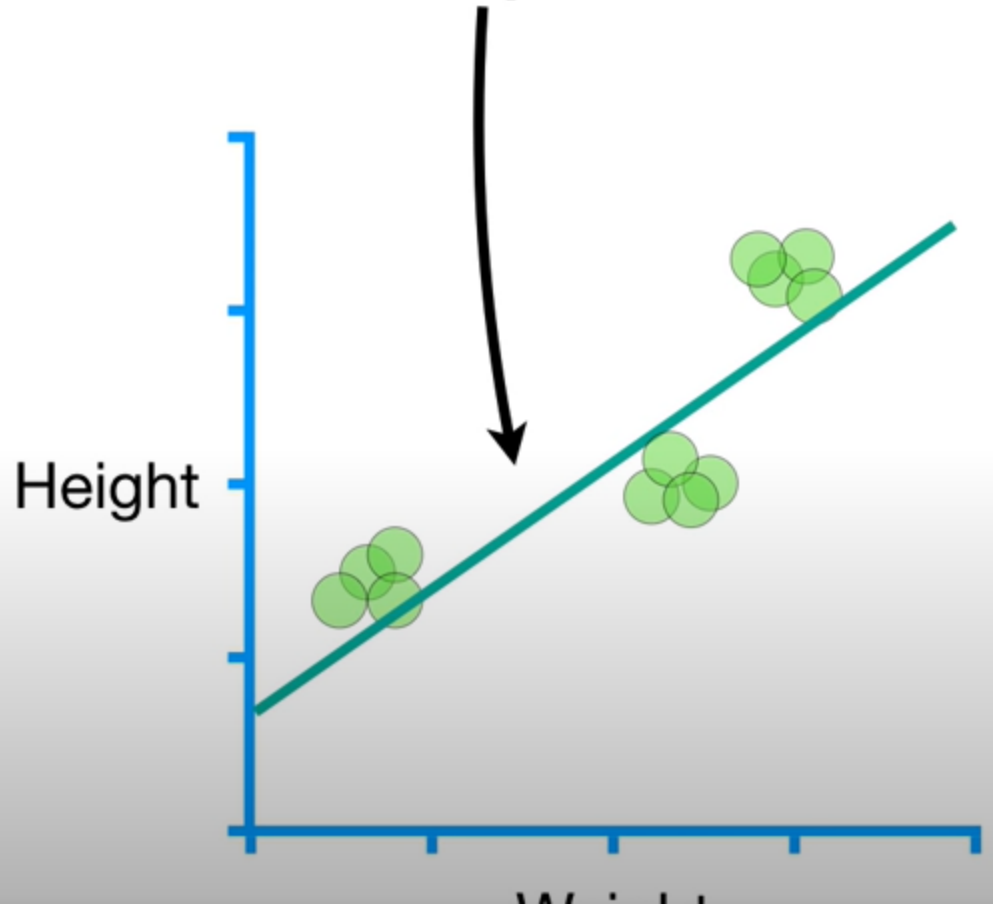


…and ultimately we end up with a line where the **intercept** = **0.85** and the **slope** = **0.68**.

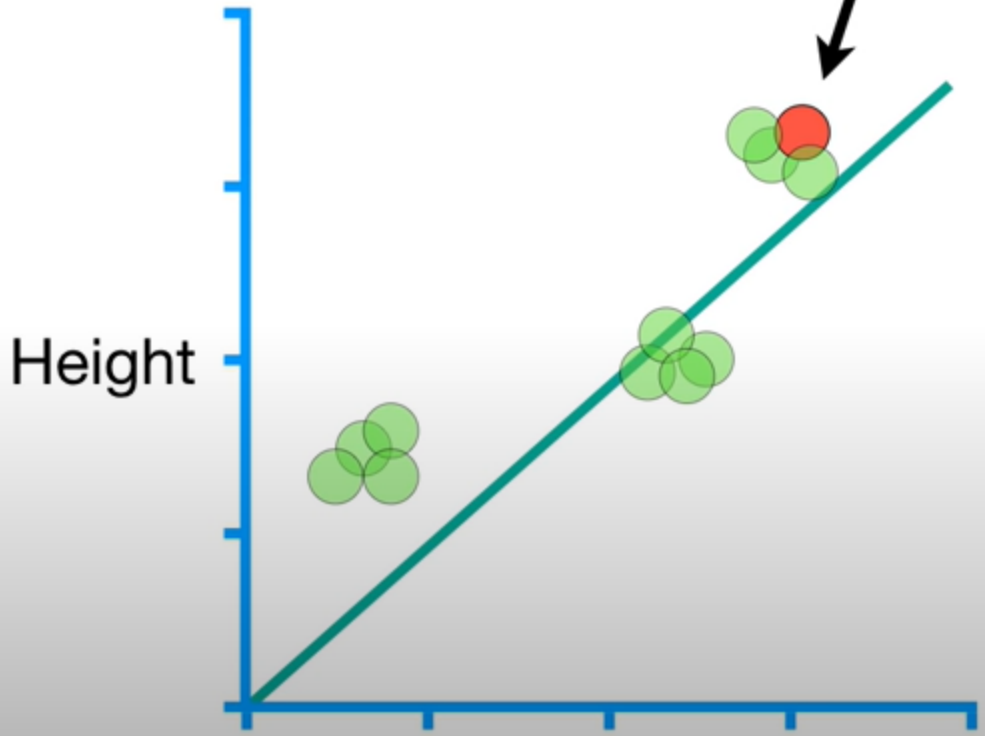


…and the least squares estimates, aka, the gold standard, gives a line where the **intercept** = **0.87** and the **slope** = **0.68**.

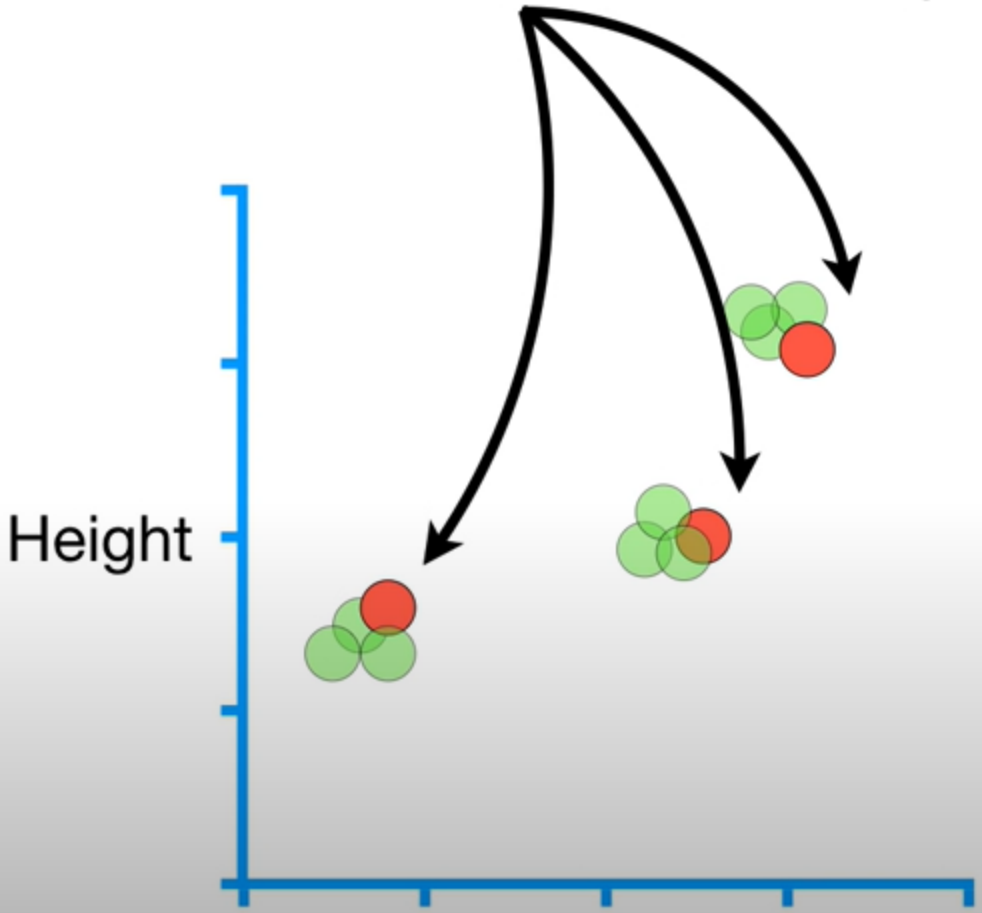
…and ultimately we end up with a line where the **intercept** = **0.85** and the **slope** = **0.68**.



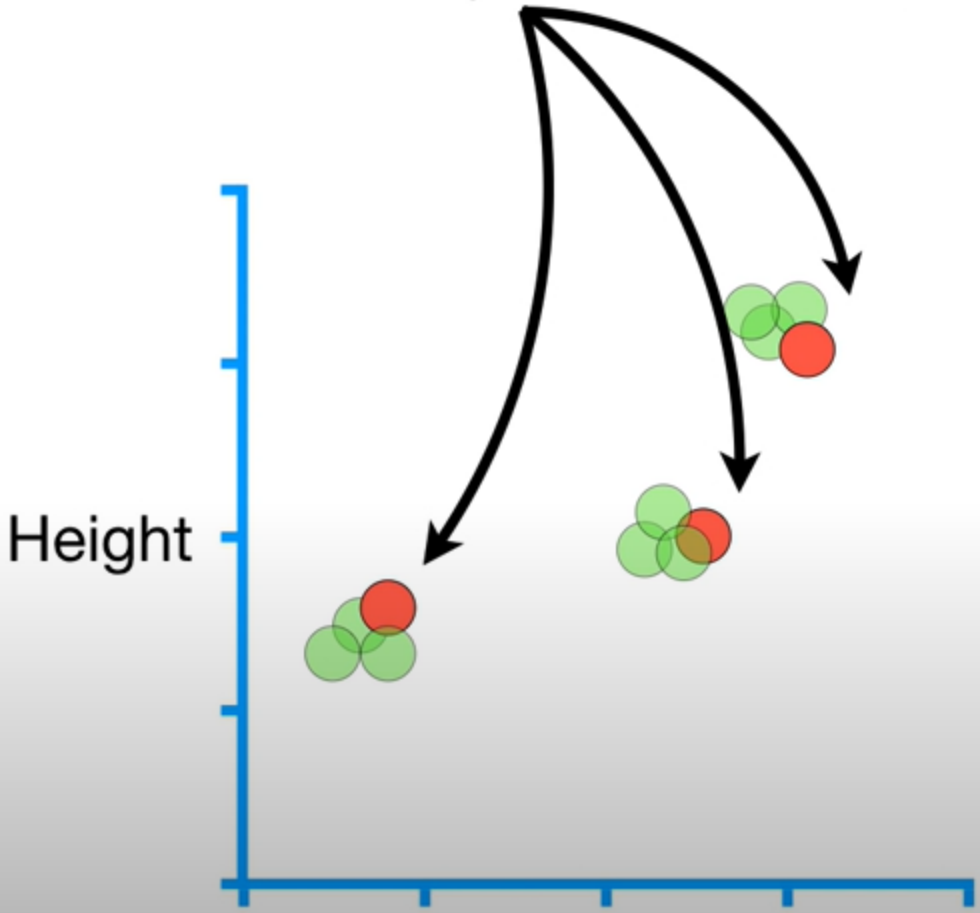
NOTE: The strict definition of **Stochastic Gradient Descent** is to only use **1** sample per step…



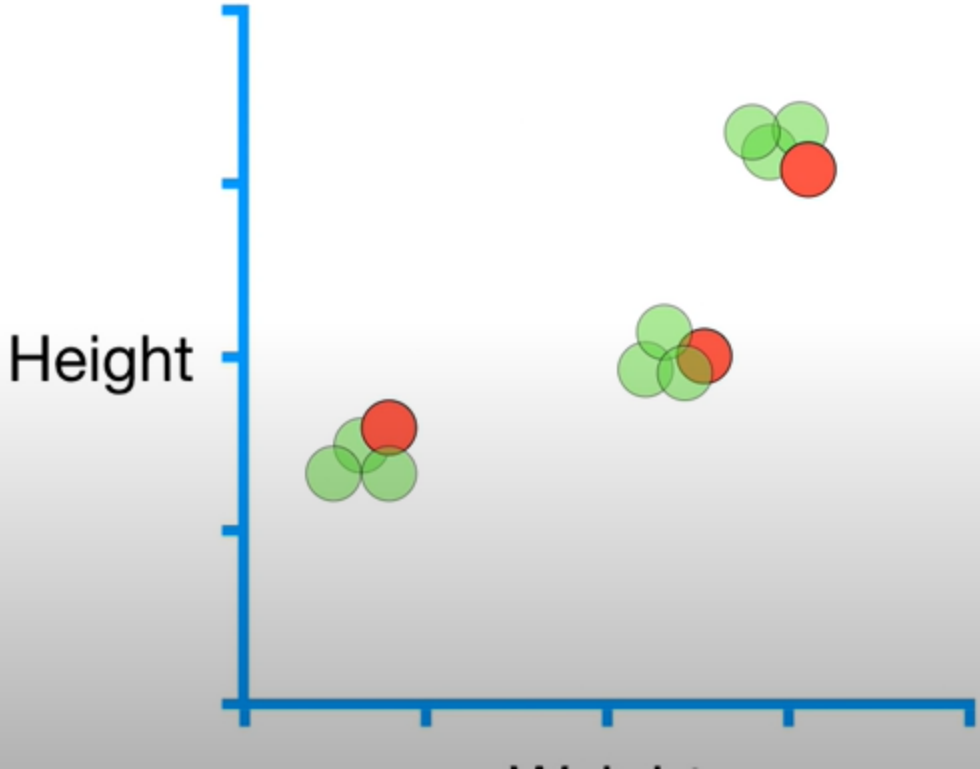
…however, it is more common
to select a small subset of data,
or **mini-batch**, for each step.



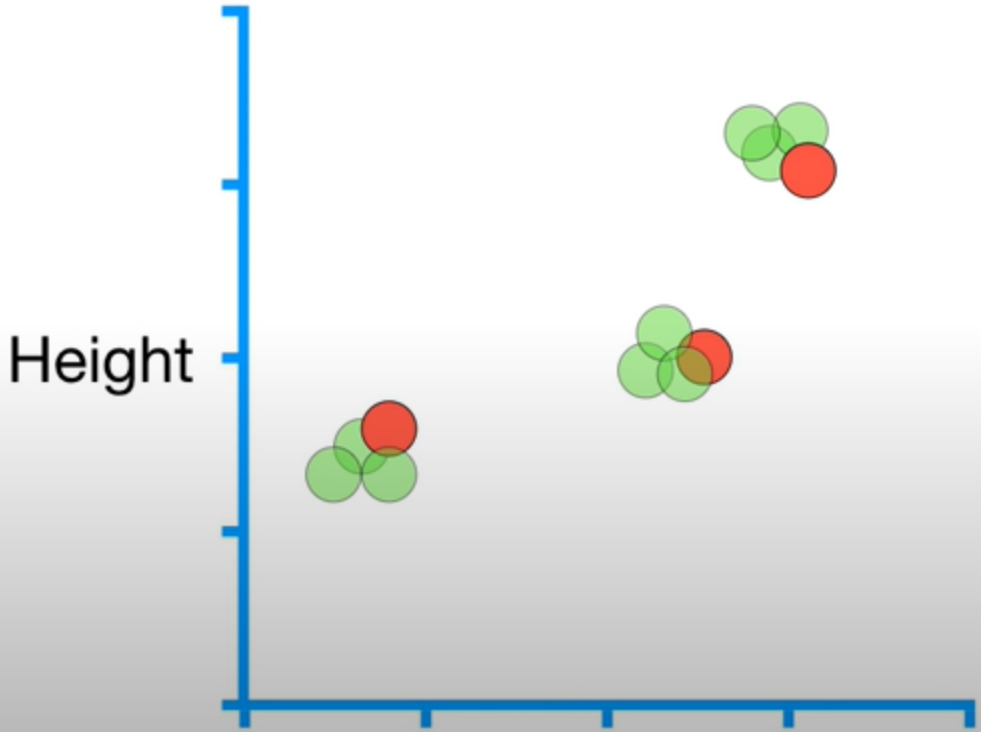
For example, we could use **3** samples per step, instead of just **1**.



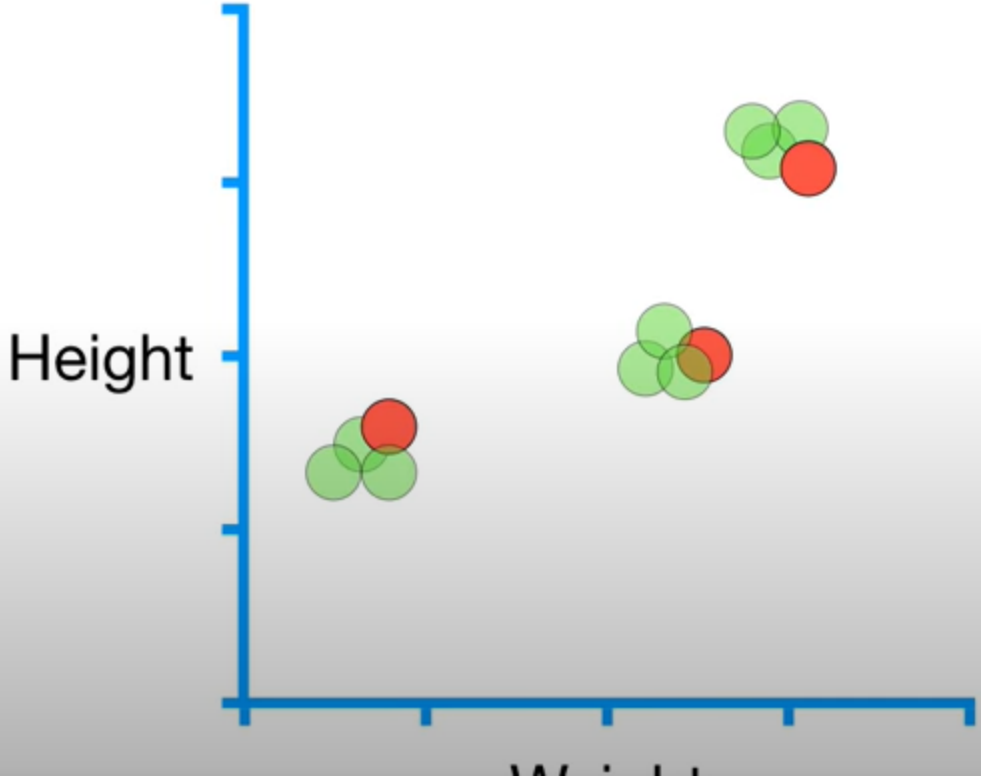
Using a **mini-batch** for each step takes the best of both worlds between using just one sample and all of of the data at each step.



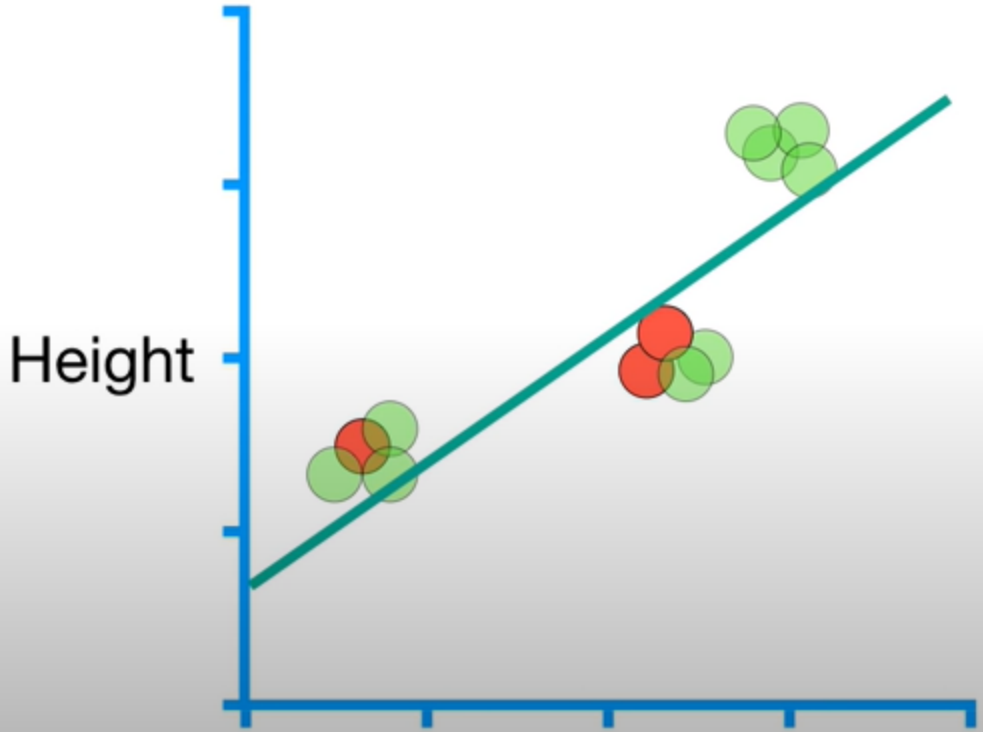
Similar to using all of the data, using a **mini-batch** can result in more stable estimates of the parameters in fewer steps…



…and like using just one sample per step, using a **mini-batch** is much faster than using all of the data.

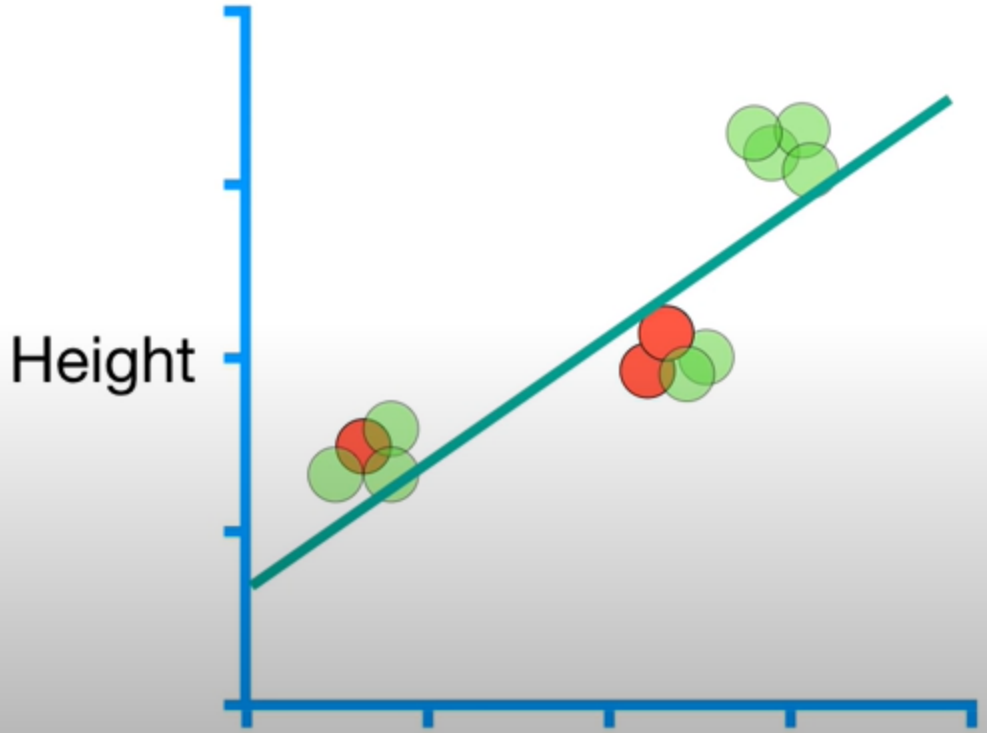


In this example, using **3** samples per step we ended up with the **intercept** = **0.86** and the **slope** = **0.68**.

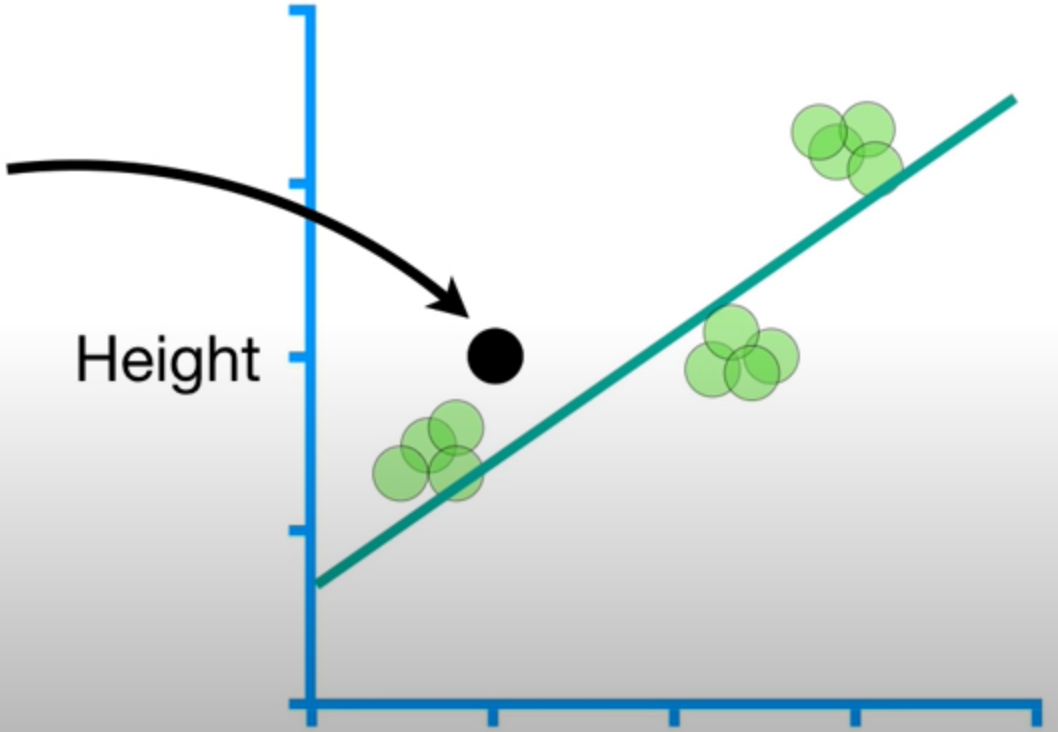


In this example, using **3** samples per step we ended up with the **intercept** = **0.86** and the **slope** = **0.68**.

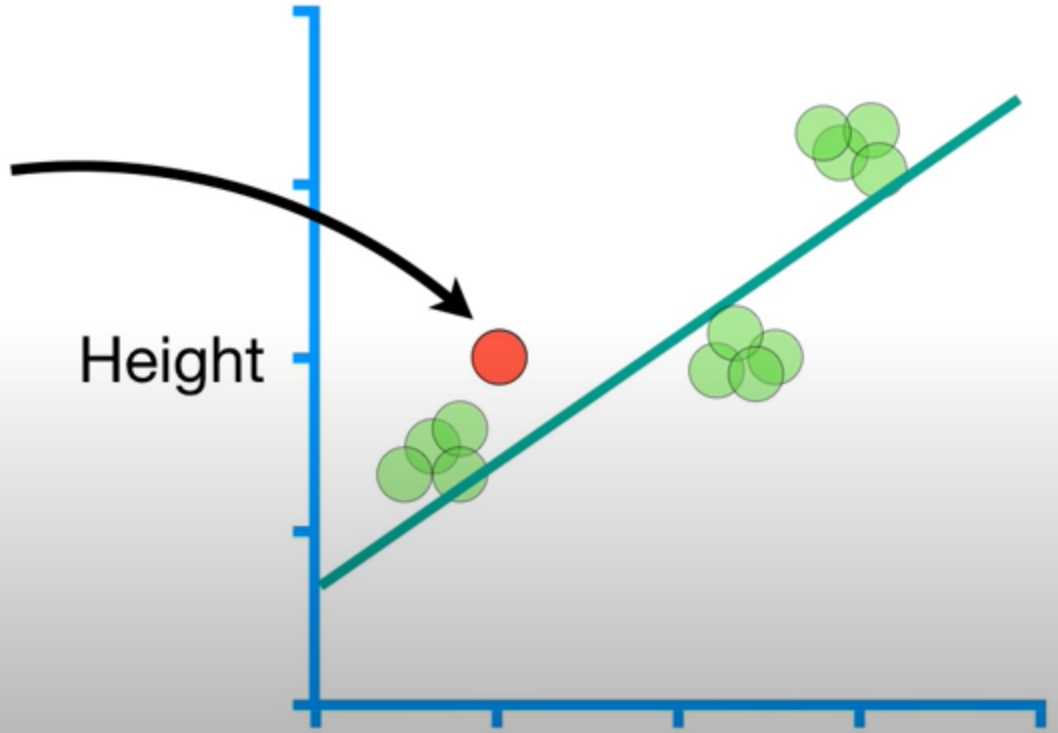
…which means that the estimate for the intercept was just a little closer to the gold standard, **0.87**, then when we used one sample and got **0.85**.



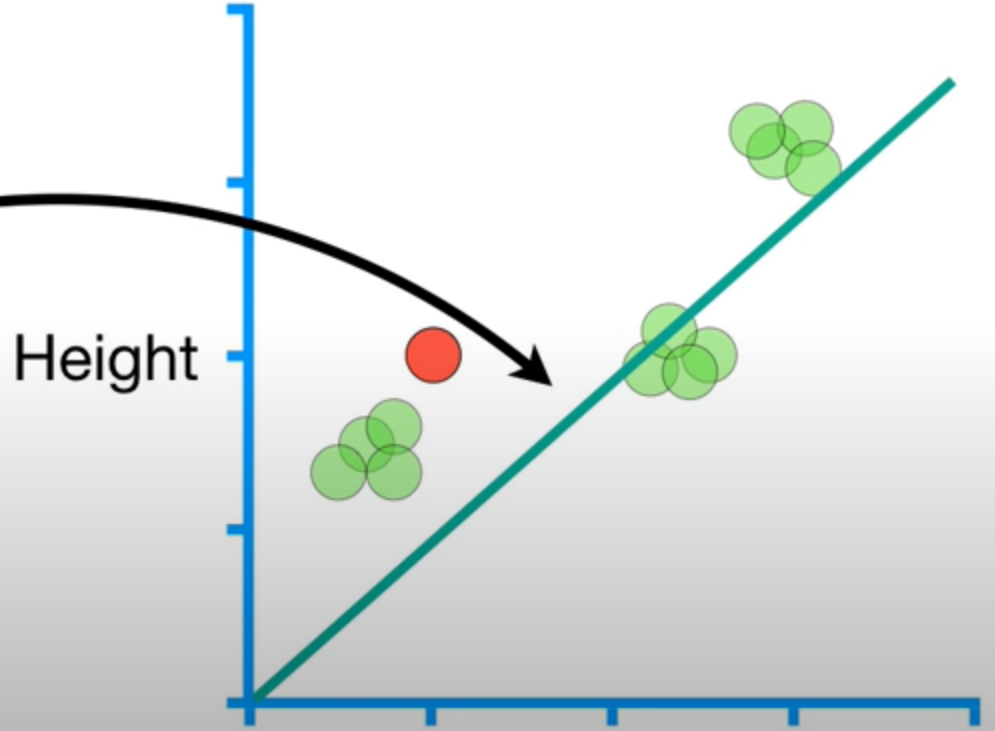
One cool thing about **Stochastic Gradient Descent** is that when we get new data…



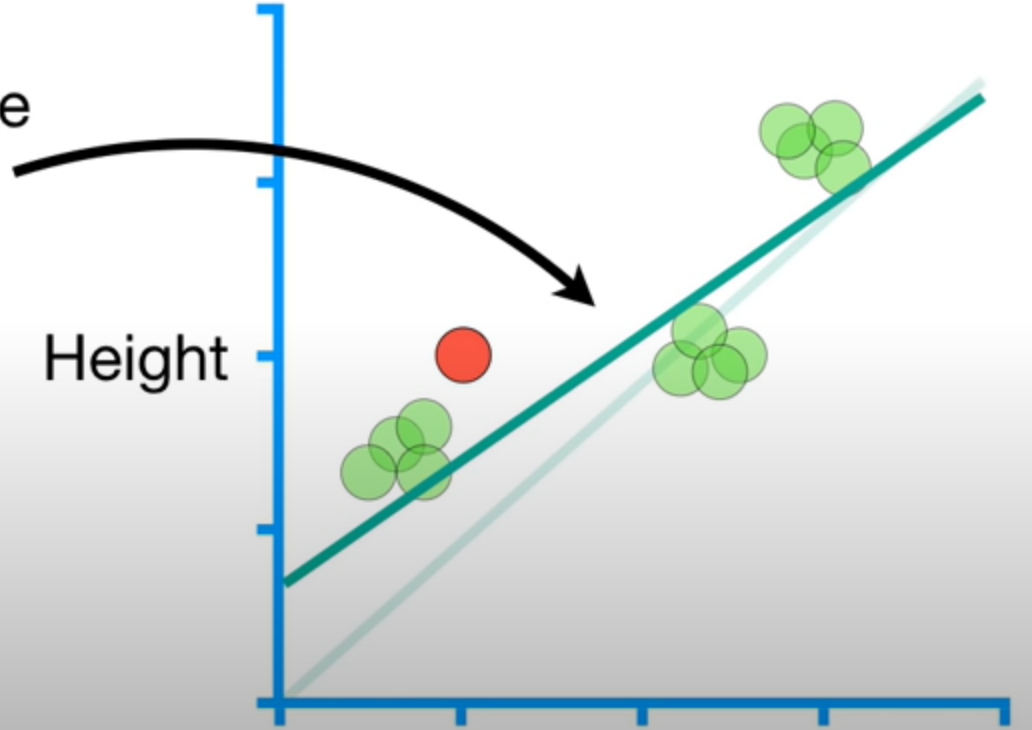
…we can easily use it to take another step for the parameter estimates without having to start from scratch.



In other words, we don't have to go all of the way back to the initial guesses for the **slope** and **intercept** and redo everything.



Instead, we pick up right where we left off and take one more step using the new sample.



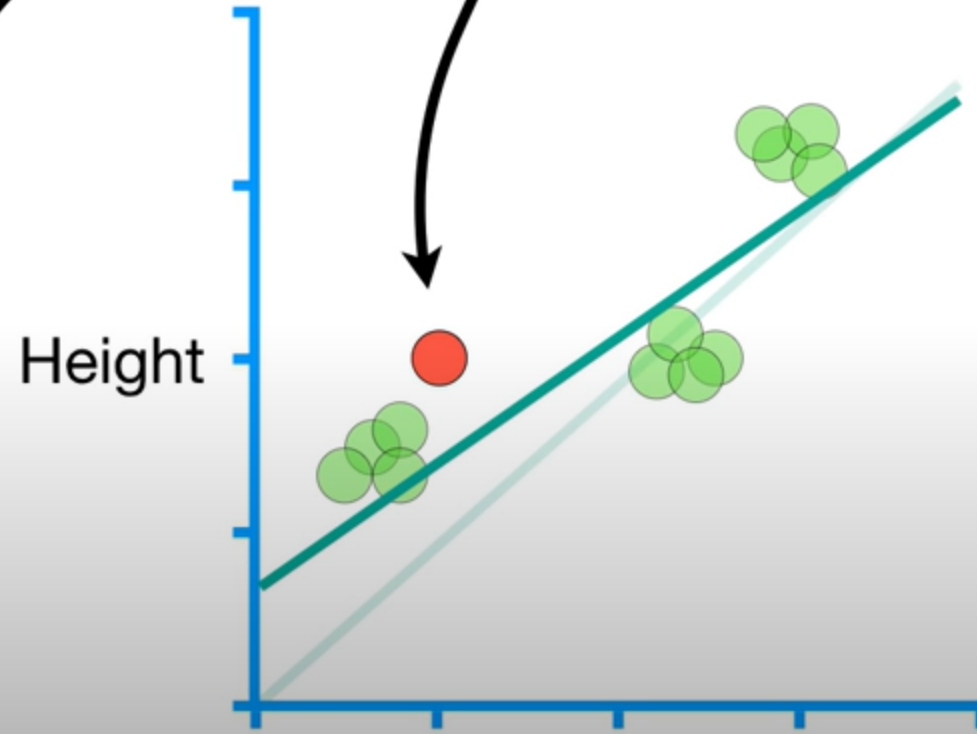
$$\frac{d}{d\ intercept}$$ Sum of squared residuals =

$$-2(\textbf{Height} - (0 + 1 \times \textbf{Weight})$$

So we plug in the **Weight** from the new sample, **1.1**…

$$\frac{d}{d\ slope}$$ Sum of squared residuals =

$$-2 \times \textbf{Weight}(\textbf{Height} - (0 + 1 \times \textbf{Weight}))$$



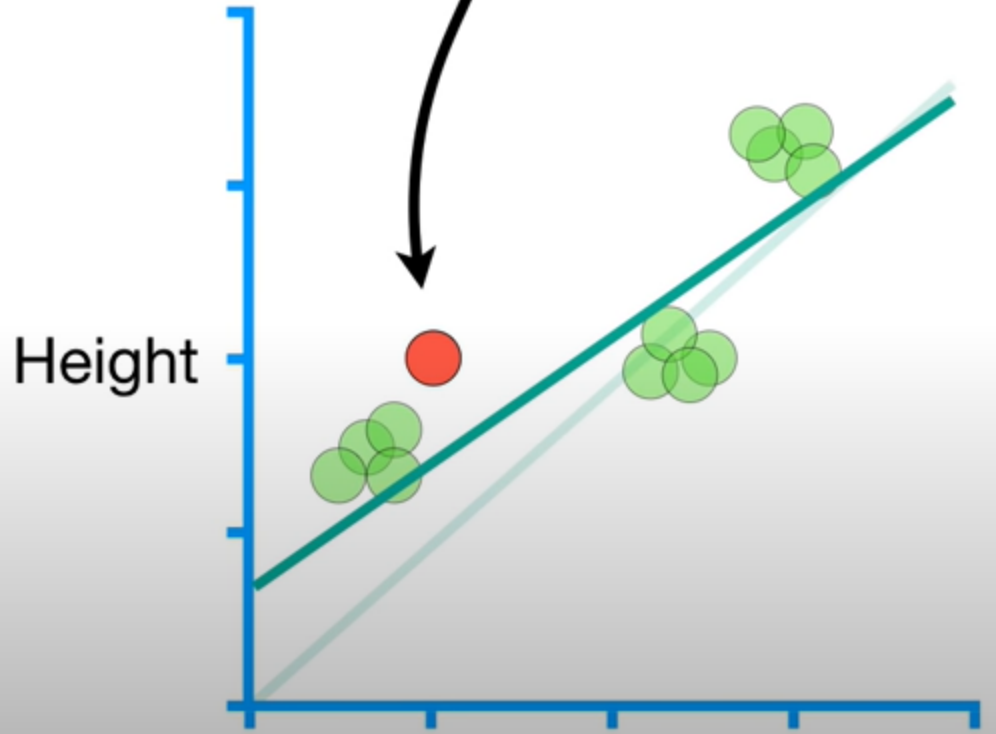
$\frac{d}{d\ intercept}$ Sum of squared residuals =

$$-2(\textbf{Height} - (0 + 1 \times \textbf{1.1})$$

So we plug in the **Weight** from the new sample, **1.1**…

$\frac{d}{d\ slope}$ Sum of squared residuals =

$$-2 \times \textbf{1.1}(\textbf{Height} - (0 + 1 \times \textbf{1.1}))$$



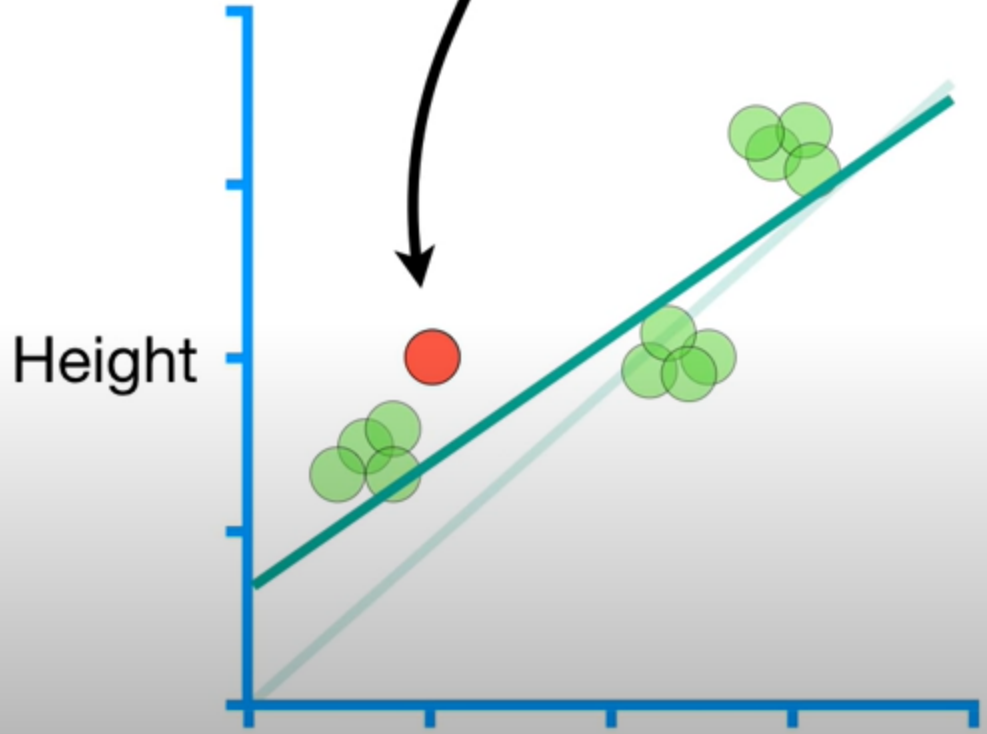
$$\frac{d}{d\ intercept}$$ Sum of squared residuals =

-2(**Height** - (0 + 1 × **1.1**)

…and the **Height**, **2**…

$$\frac{d}{d\ slope}$$ Sum of squared residuals =

-2 × **1.1**(**Height** - (0 + 1 × **1.1**))



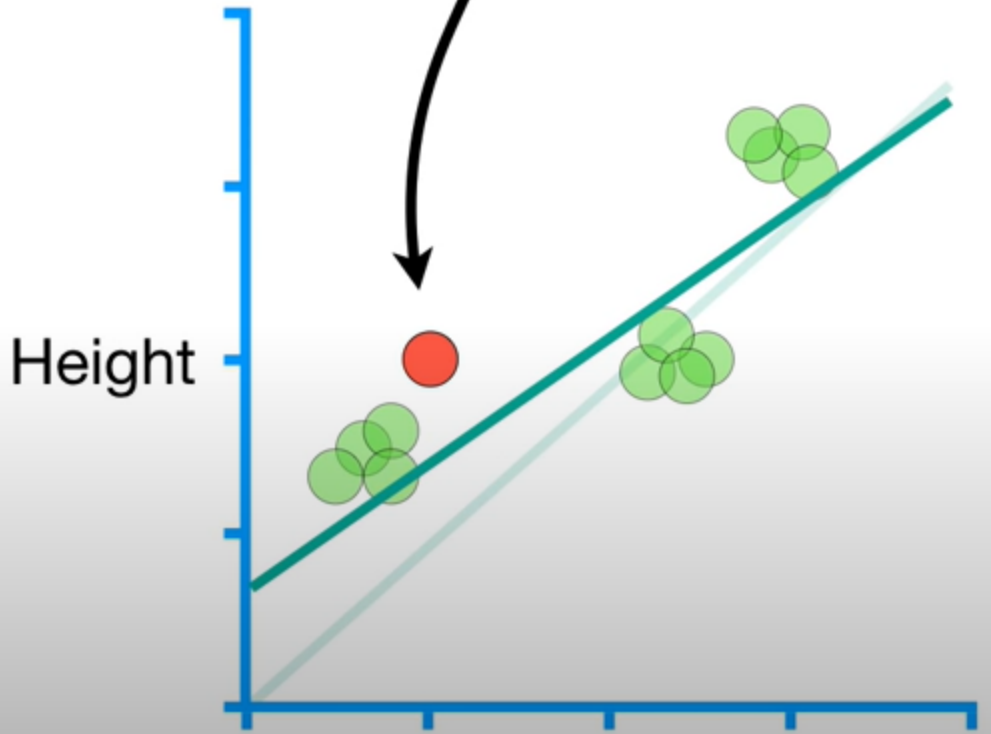
$$\frac{d}{d\ intercept}$$ Sum of squared residuals =

-2(**2** - (0 + 1 × **1.1**)

…and the **Height**, **2**…

$$\frac{d}{d\ slope}$$ Sum of squared residuals =

-2 × **1.1**(**2** - (0 + 1 × **1.1**))



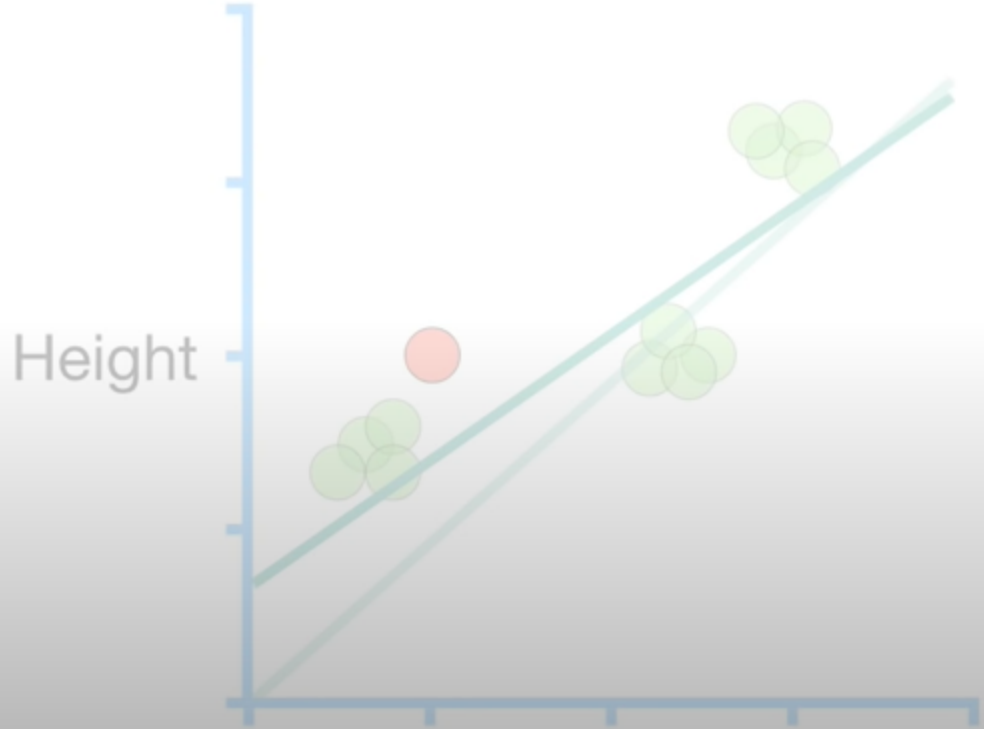
$\frac{d}{d\ intercept}$ Sum of squared residuals =

$-2(\mathbf{2} - (0 + 1 \times \mathbf{1.1}) = -1.8$

…do the math…

$\frac{d}{d\ slope}$ Sum of squared residuals =

$-2 \times \mathbf{1.1}(\mathbf{2} - (0 + 1 \times \mathbf{1.1})) = -1.98$



$\frac{d}{d\ intercept}$ Sum of squared residuals =

$-2(\mathbf{2} - (0 + 1 \times \mathbf{1.1}) =$ -1.8

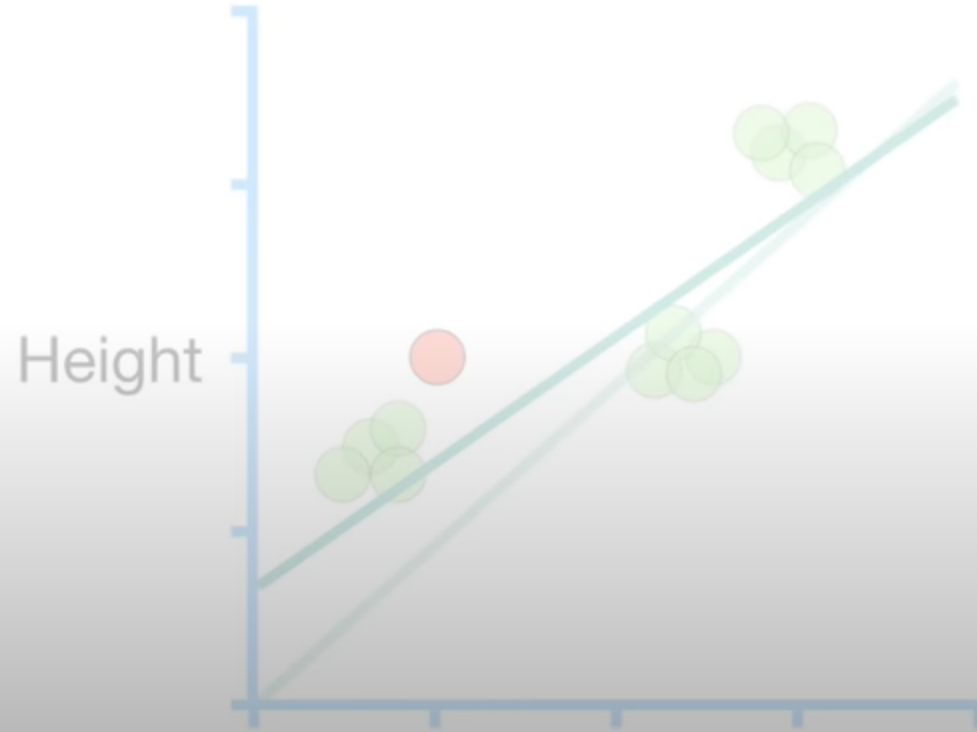
Step Size$_{\text{Intercept}}$ = **Slope** x **Learning Rate**

…plug in the slopes…

Step Size$_{\text{Slope}}$ = **Slope** x **Learning Rate**

$\frac{d}{d\ slope}$ Sum of squared residuals =

$-2 \times \mathbf{1.1}(\mathbf{2} - (0 + 1 \times \mathbf{1.1})) =$ -1.98



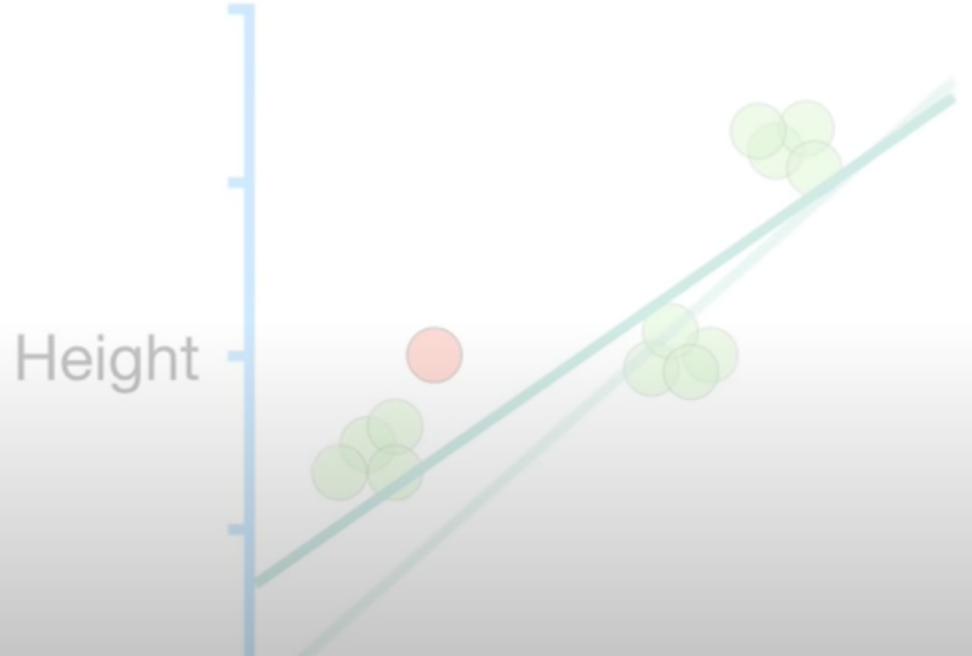
$\frac{d}{d\ intercept}$ Sum of squared residuals = $-2(\mathbf{2} - (0 + 1 \times \mathbf{1.1}) = -1.8$

Step Size$_{\text{Intercept}}$ = -1.8 x **Learning Rate**

…plug in the slopes…

Step Size$_{\text{Slope}}$ = -1.98 x **Learning Rate**

$\frac{d}{d\ slope}$ Sum of squared residuals = $-2 \times \mathbf{1.1}(\mathbf{2} - (0 + 1 \times \mathbf{1.1})) = -1.98$



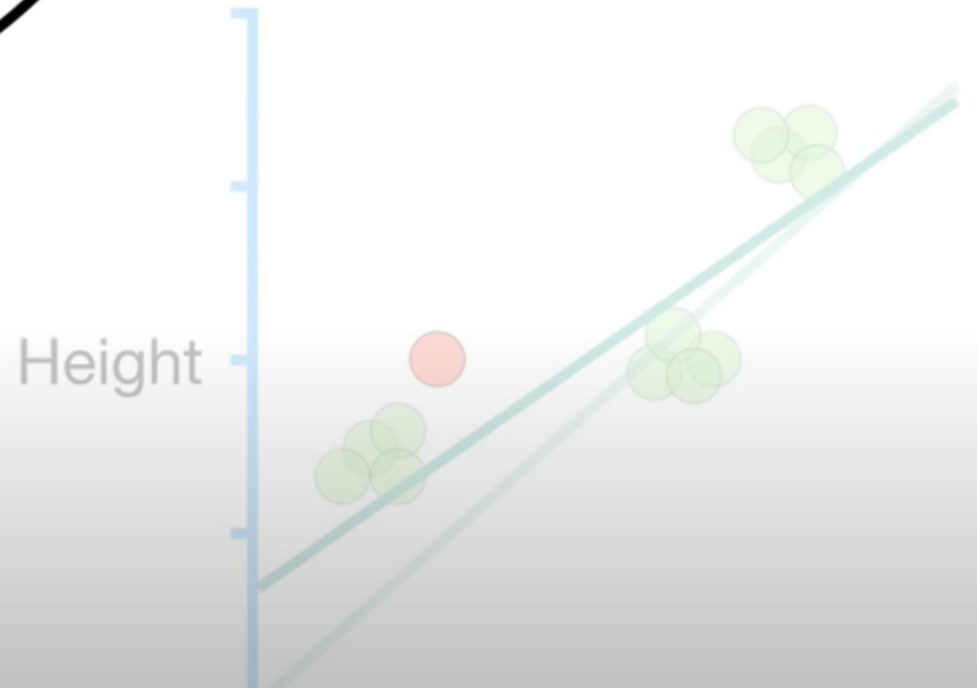
…then multiply by the
Learning Rate, 0.01…

Step Size$_{\text{Intercept}}$ = -1.8 × **Learning Rate**

Step Size$_{\text{Slope}}$ = -1.98 × **Learning Rate**

$\frac{d}{d\ slope}$ Sum of squared residuals =

-2 × **1.1**(**2** - (0 + 1 × **1.1**)) = -1.98



$\frac{d}{d\ intercept}$ Sum of squared residuals =

$-2(\mathbf{2} - (0 + 1 \times \mathbf{1.1}) = -1.8$

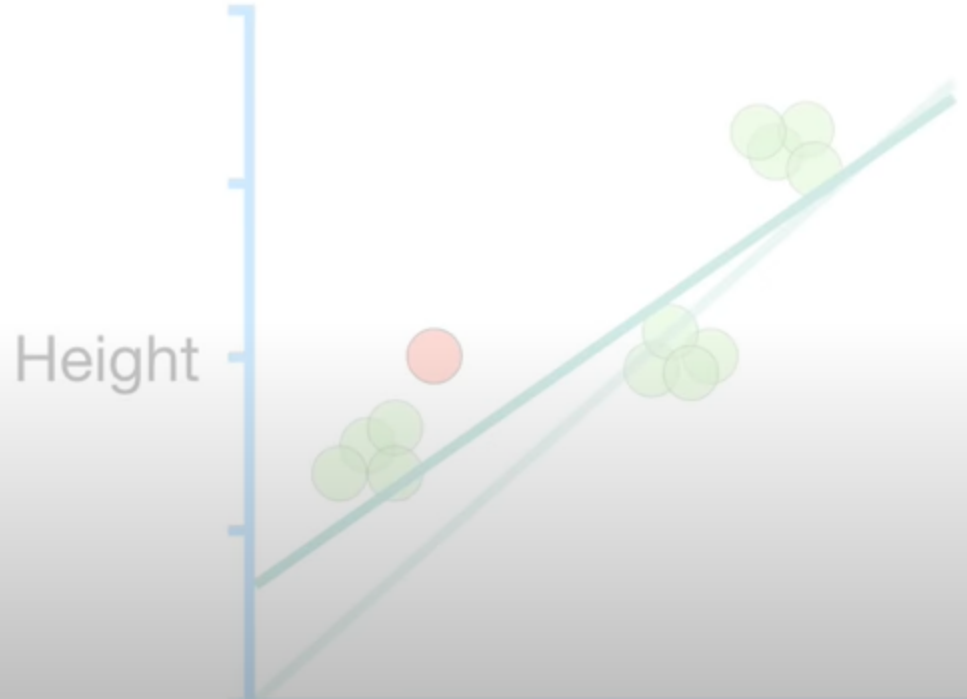
…then multiply by the **Learning Rate, 0.01**…

Step Size$_{Intercept}$ = -1.8 x 0.01

Step Size$_{Slope}$ = -1.98 x 0.01

$\frac{d}{d\ slope}$ Sum of squared residuals =

$-2 \times \mathbf{1.1}(\mathbf{2} - (0 + 1 \times \mathbf{1.1})) = -1.98$



$$\frac{d}{d\ intercept}$$ Sum of squared residuals =

$$-2(\mathbf{2} - (0 + 1 \times \mathbf{1.1}) = -1.8$$

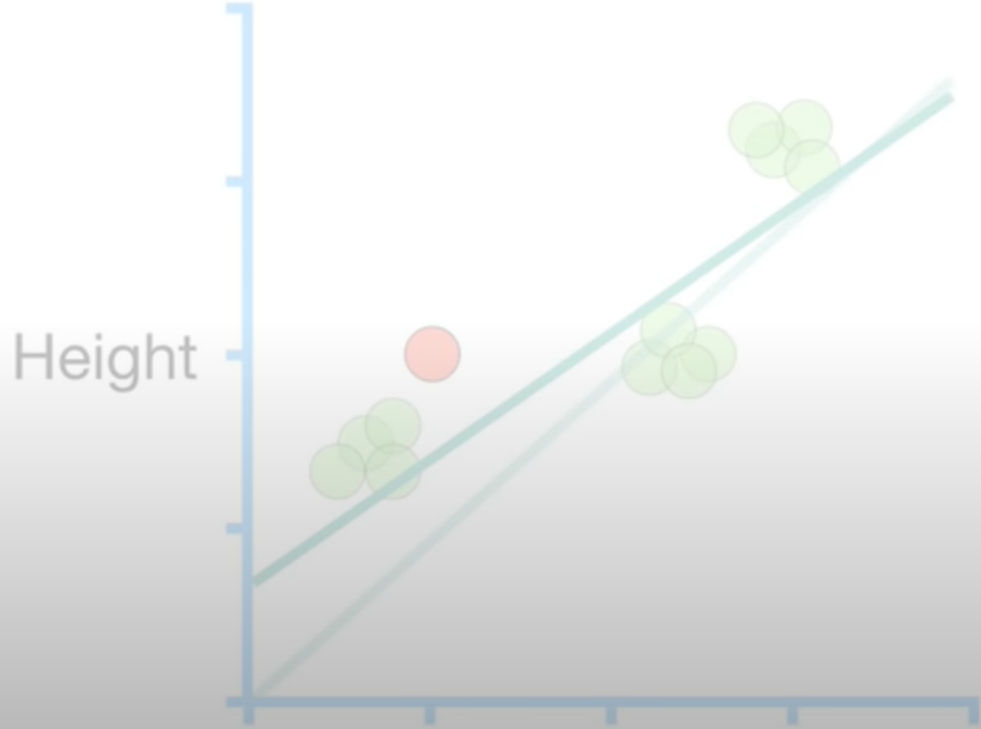
…do the math…

Step Size$_{Intercept}$ = -1.8 x 0.01

Step Size$_{Slope}$ = -1.98 x 0.01

$$\frac{d}{d\ slope}$$ Sum of squared residuals =

$$-2 \times \mathbf{1.1}(\mathbf{2} - (0 + 1 \times \mathbf{1.1})) = -1.98$$



$\frac{d}{d\ intercept}$ Sum of squared residuals =

$-2(\mathbf{2} - (0 + 1 \times \mathbf{1.1}) = -1.8$

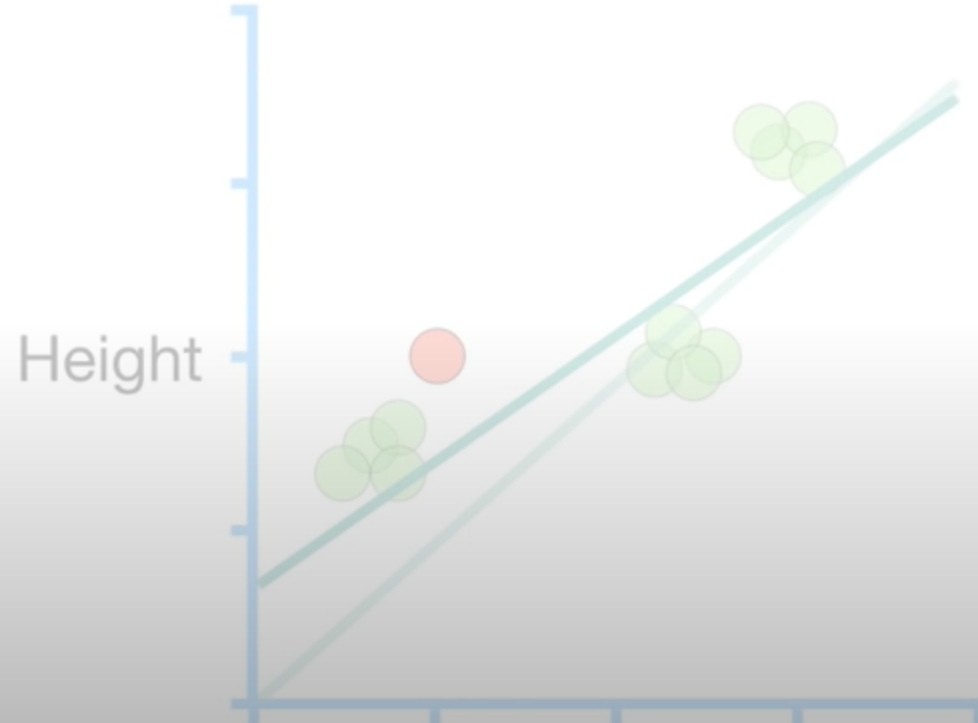
…do the math…

Step Size$_{Intercept}$ = -1.8 x 0.01 = **-0.018**

Step Size$_{Slope}$ = -1.98 x 0.01 = **-0.02**

$\frac{d}{d\ slope}$ Sum of squared residuals =

$-2 \times \mathbf{1.1}(\mathbf{2} - (0 + 1 \times \mathbf{1.1})) = -1.98$

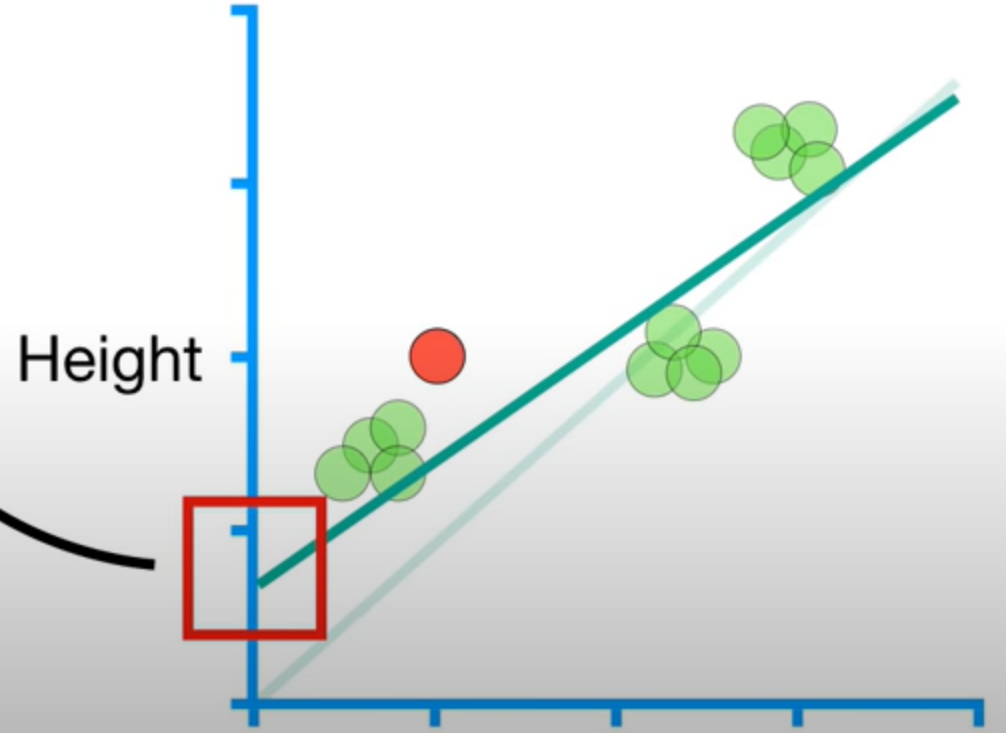


New Intercept = **Old Intercept** - **Step Size**

Step Size$_{\text{Intercept}}$ = -1.8 x 0.01 = **-0.018**

Step Size$_{\text{Slope}}$ = -1.98 x 0.01 = **-0.02**

…calculate the new **intercept,** not from the initial guess, but from the most recent estimate…

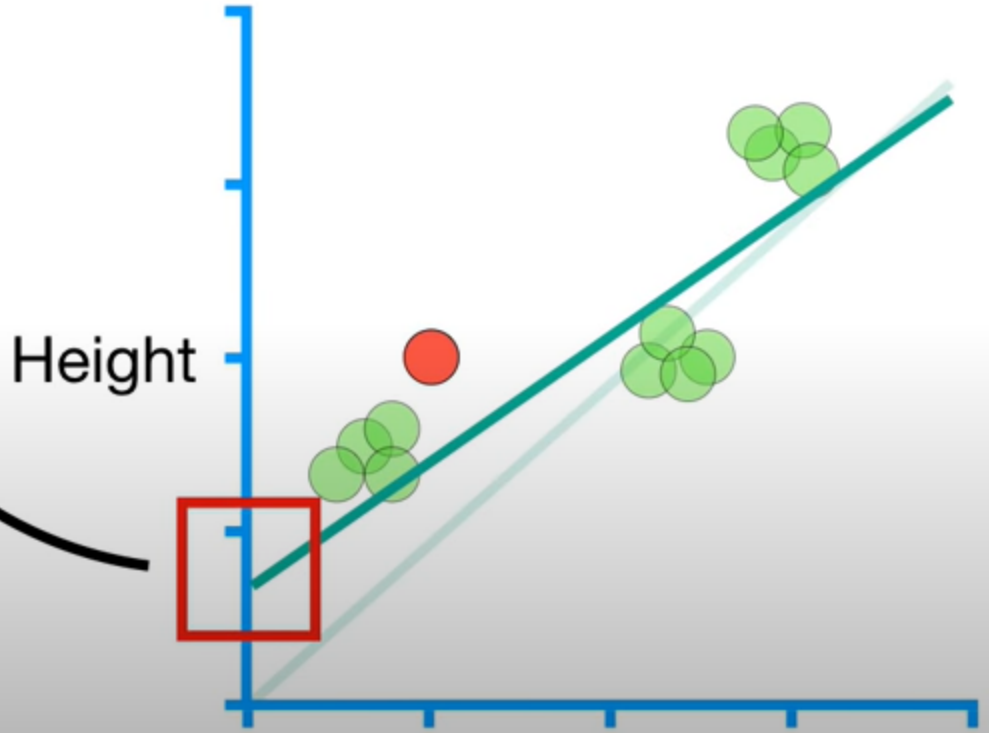


New Intercept = 0.86 - **Step Size**

Step Size$_{\text{Intercept}}$ = -1.8 x 0.01 = **-0.018**

Step Size$_{\text{Slope}}$ = -1.98 x 0.01 = **-0.02**

…calculate the new **intercept,** not from the initial guess, but from the most recent estimate…

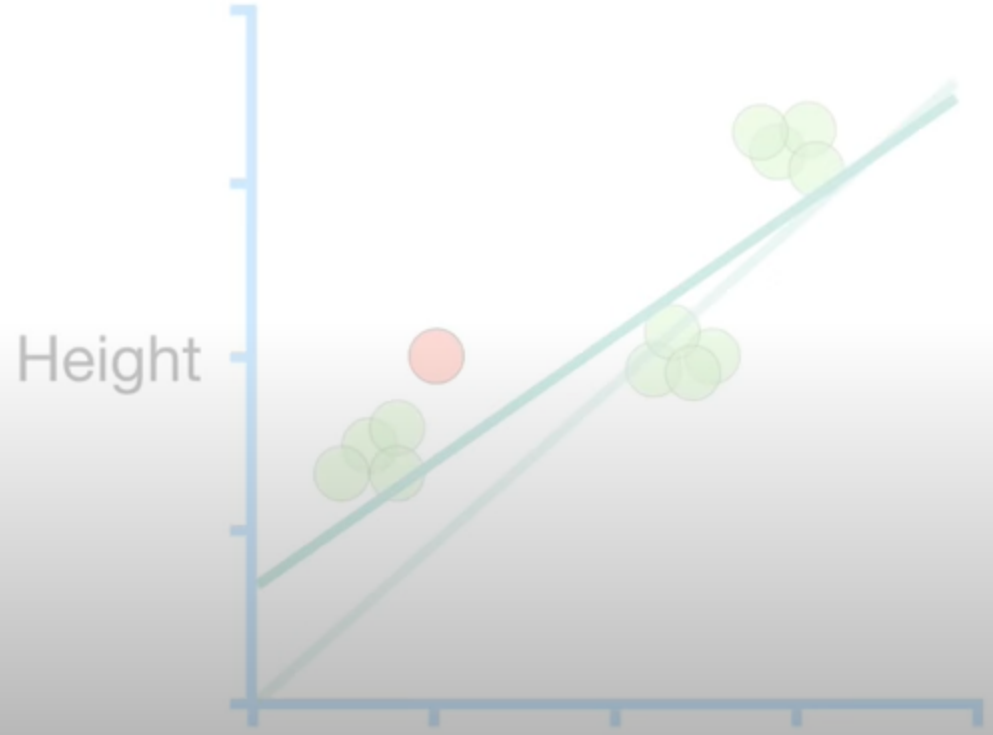


New Intercept = 0.86 - **Step Size**

Step Size$_{\text{Intercept}}$ = -1.8 × 0.01 = **-0.018**

Step Size$_{\text{Slope}}$ = -1.98 × 0.01 = **-0.02**

…calculate the new **intercept,** not from the initial guess, but from the most recent estimate…

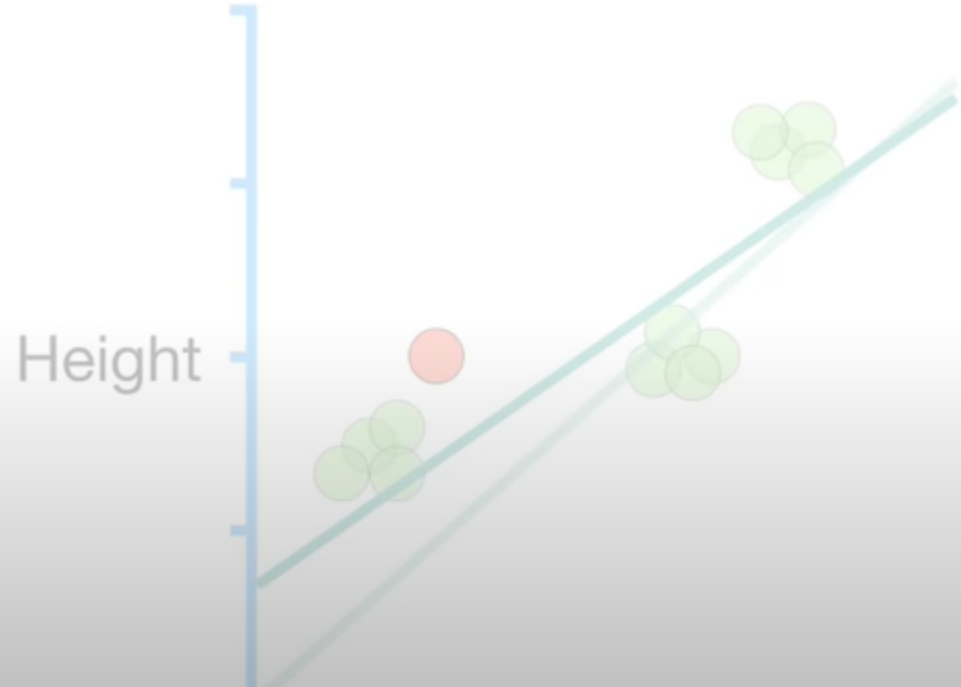


New Intercept = 0.86 - -0.018

Step Size$_{\text{Intercept}}$ = -1.8 x 0.01 = **-0.018**

Step Size$_{\text{Slope}}$ = -1.98 x 0.01 = **-0.02**

…calculate the new **intercept,** not from the initial guess, but from the most recent estimate…



New Intercept = 0.86 - -0.018 = 0.878

Step Size$_{\text{Intercept}}$ = -1.8 x 0.01 = **-0.018**

Step Size$_{\text{Slope}}$ = -1.98 x 0.01 = **-0.02**

…calculate the new **intercept,** not from the initial guess, but from the most recent estimate…

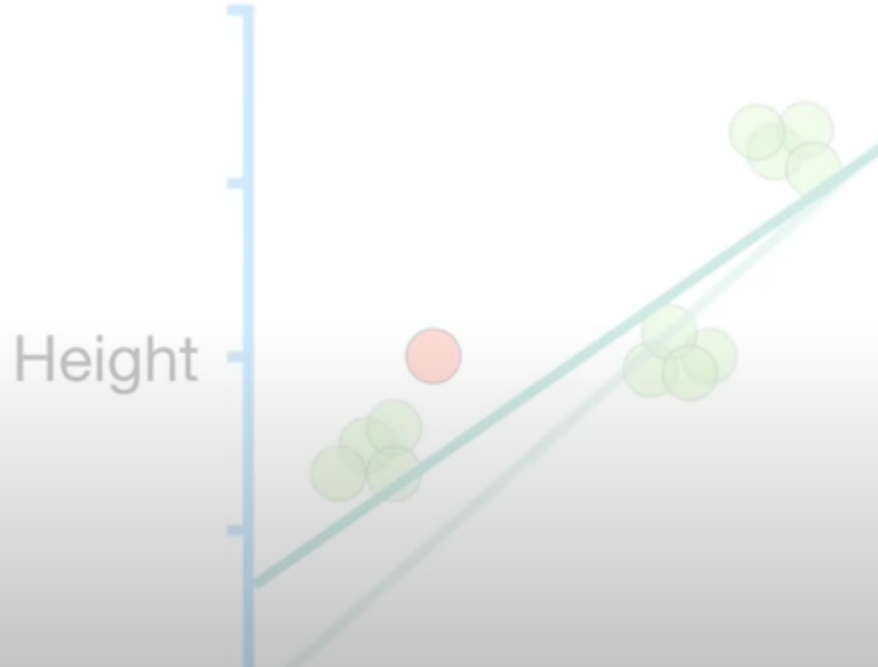
New Intercept = 0.86 - -0.018 = 0.878

Step Size$_{\text{Intercept}}$ = -1.8 x 0.01 = **-0.018**

Step Size$_{\text{Slope}}$ = -1.98 x 0.01 = **-0.02**

New Slope = **Old Slope** - **Step Size**

…and calculate the new **slope** from the most recent estimate…



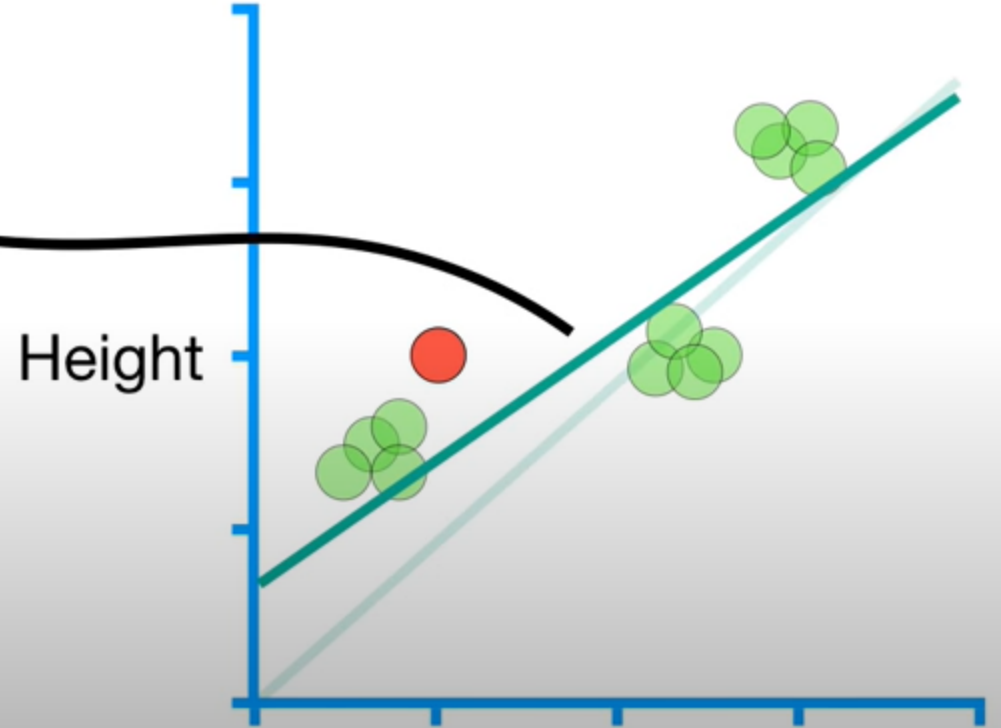
New Intercept = 0.86 - -0.018 = 0.878

Step Size$_{\text{Intercept}}$ = -1.8 x 0.01 = **-0.018**

…and calculate the new **slope** from the most recent estimate…

Step Size$_{\text{Slope}}$ = -1.98 x 0.01 = **-0.02**

New Slope = **Old Slope** - **Step Size**



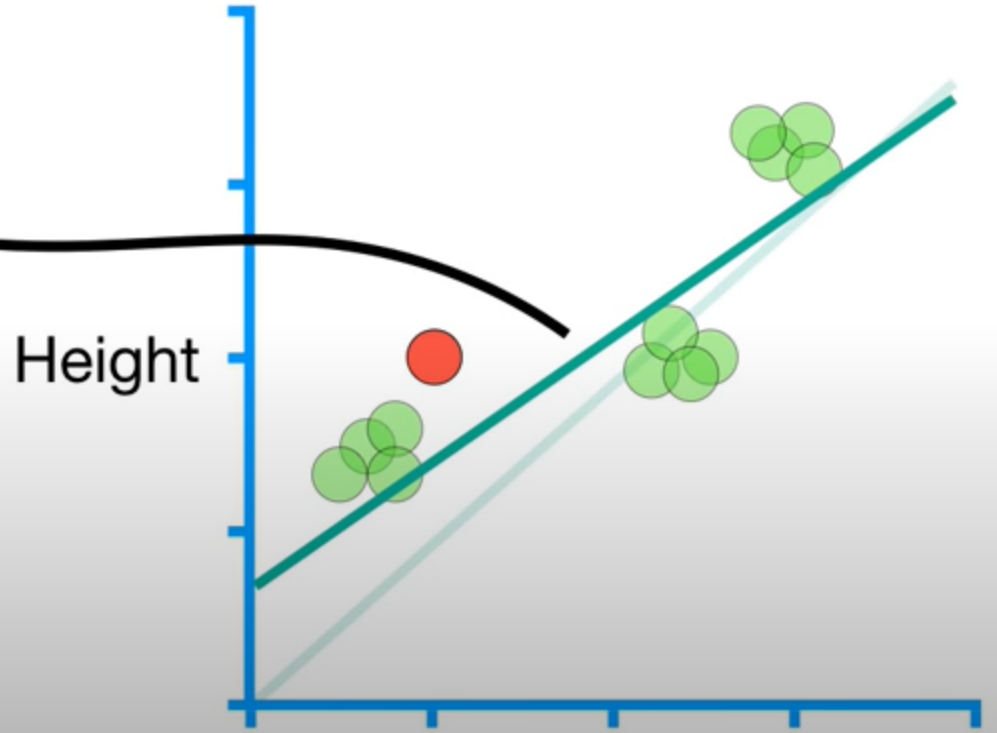
New Intercept = 0.86 - -0.018 = 0.878

Step Size$_{\text{Intercept}}$ = -1.8 × 0.01 = **-0.018**

…and calculate the new **slope** from the most recent estimate…

Step Size$_{\text{Slope}}$ = -1.98 × 0.01 = **-0.02**

New Slope = 0.68 - **Step Size**



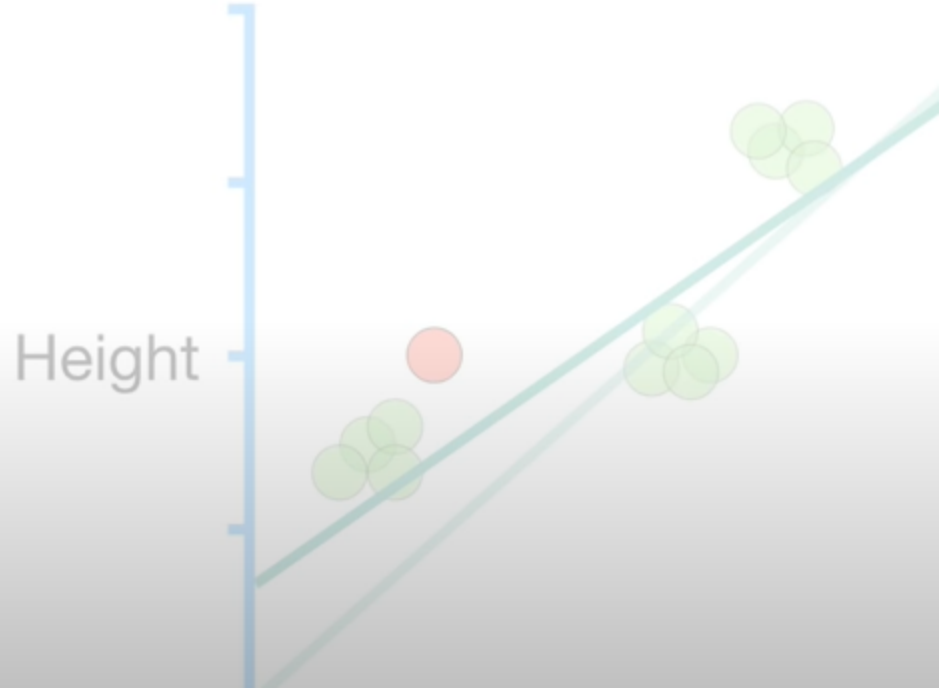
New Intercept = 0.86 - -0.018 = 0.878

Step Size$_{\text{Intercept}}$ = -1.8 × 0.01 = **-0.018**

Step Size$_{\text{Slope}}$ = -1.98 × 0.01 = **-0.02**

New Slope = 0.68 - **Step Size**

…and calculate the new **slope** from the most recent estimate…



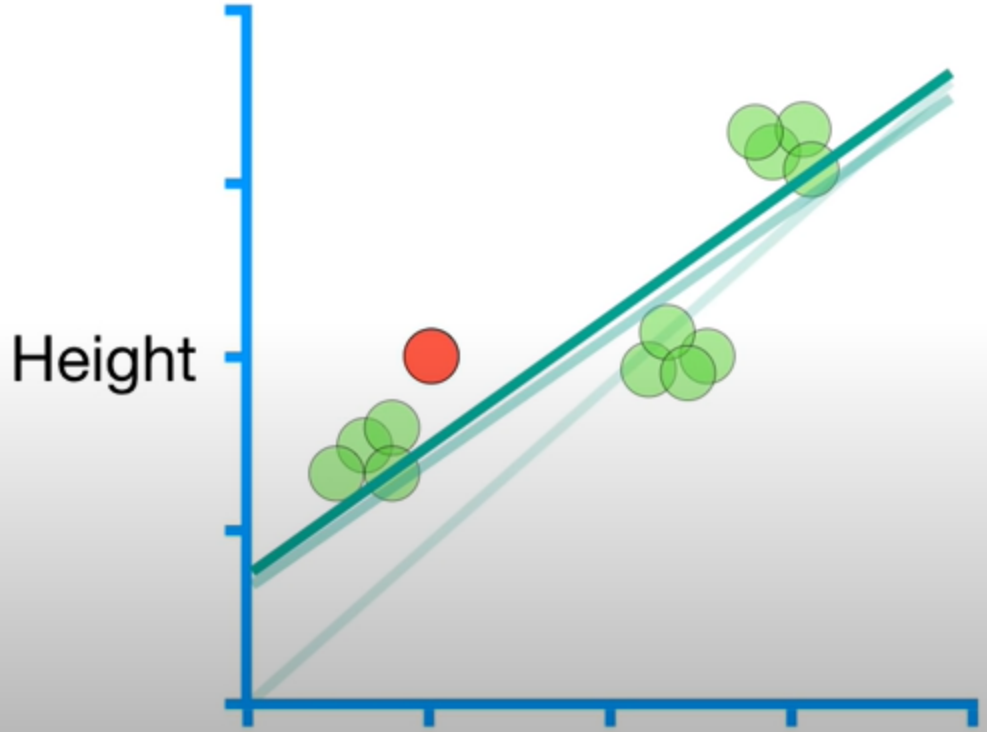
New Intercept = 0.86 - -0.018 = 0.878

$\textbf{Step Size}_{\text{Intercept}} = -1.8 \times 0.01 = \textbf{-0.018}$

$\textbf{Step Size}_{\text{Slope}} = -1.98 \times 0.01 = \textbf{-0.02}$

New Slope = 0.68 - -0.02 = 0.7

…and the new line has **intercept** = **0.878** and **slope** = **0.07**.



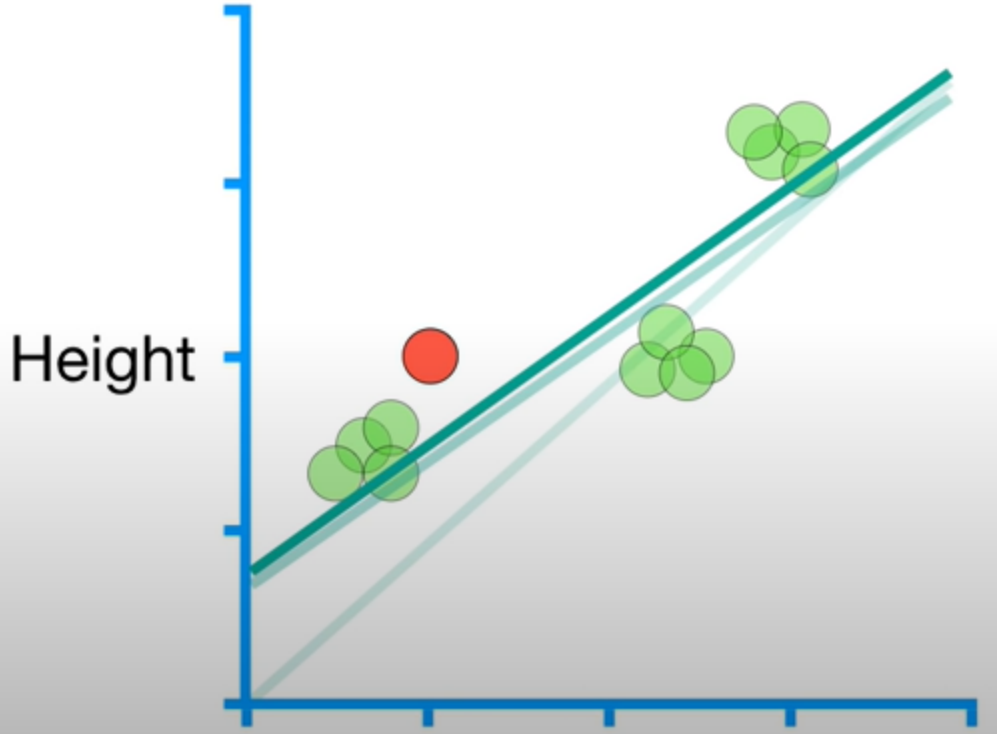
New Intercept = 0.86 - -0.018 = 0.878

Step Size$_{\text{Intercept}}$ = -1.8 × 0.01 = **-0.018**

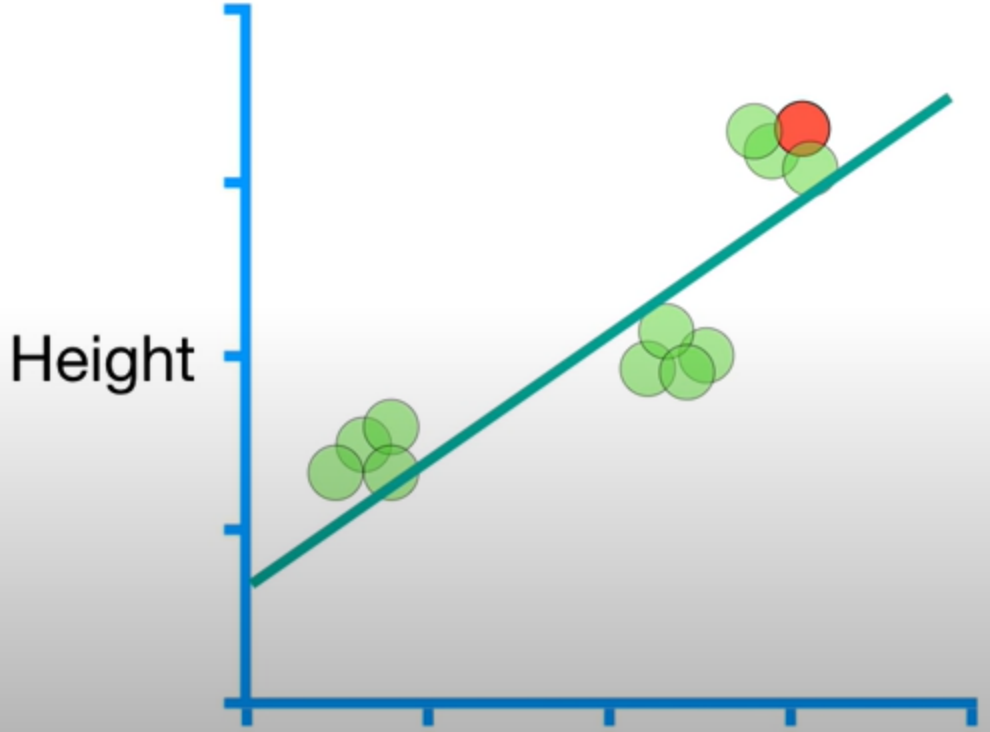
Step Size$_{\text{Slope}}$ = -1.98 × 0.01 = **-0.02**

New Slope = 0.68 - -0.02 = 0.7

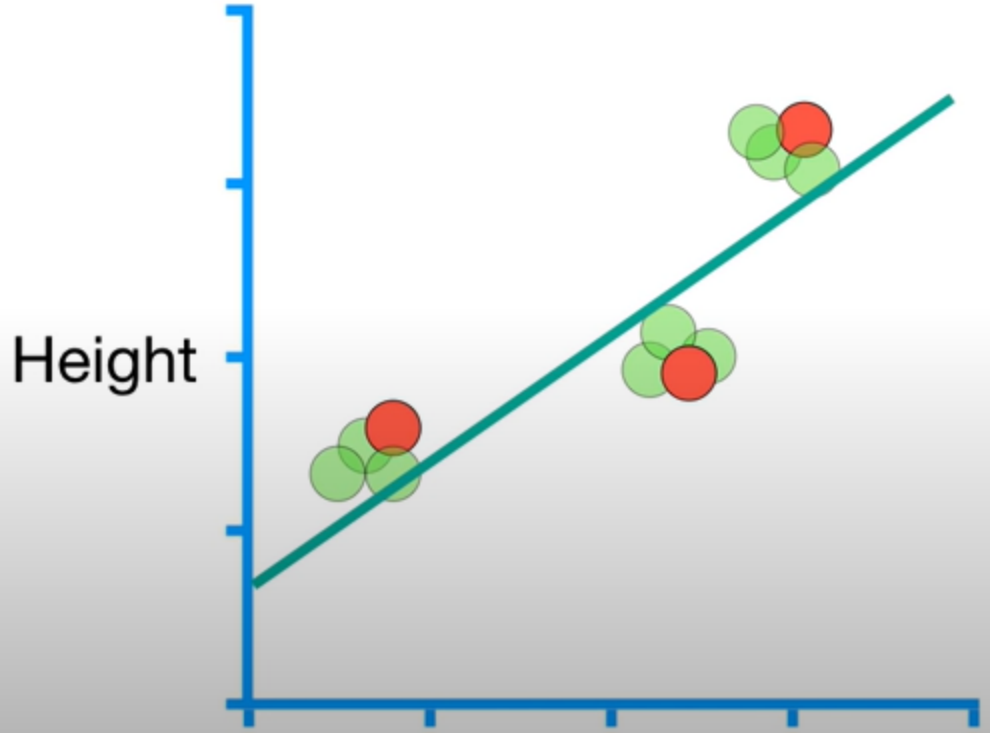
We updated the parameters for the line with just the new data.



Stochastic Gradient Descent is just like regular **Gradient Descent**, except it only looks at one sample per step…



…or a small subset, or
mini-batch, for each step.



$\frac{d}{d\,gene1}$ Loss Function()

$\frac{d}{d\,gene2}$ Loss Function()

$\frac{d}{d\,gene3}$ Loss Function()

$\frac{d}{d\,gene4}$ Loss Function()

$\frac{d}{}$ Loss Function()

Stochastic Gradient Descent is great when we have tons of data and a lot of parameters.

$\frac{d}{d\ gene1}$ Loss Function()

$\frac{d}{d\ gene2}$ Loss Function()

$\frac{d}{d\ gene3}$ Loss Function()

$\frac{d}{d\ gene4}$ Loss Function()

$\frac{d}{d\ gene5}$ Loss Function()

$\frac{d}{d\ gene6}$ Loss Function()

$\frac{d}{d\ gene7}$ Loss Function()

Stochastic Gradient Descent is great when we have tons of data and a lot of parameters.

In these situations, regular **Gradient Descent** may not be computationally feasible.

And it's cool that we can easily update the parameters when new data shows up.

